

Servista

Broad Range of SERVICES

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Under the Supervision of:

Prof. Abu Zar Tamimi

Bachelor of Science in *Information Technology*

Session (2022-2026).

Department of Information Technology

**A Project represented to
University of the Punjab, Lahore.**



STATEMENT OF SUBMISSION

This is to certify that the following students have successfully completed the final year project titled “**Servista**” at **Govt. Jinnah Islamia College, Sialkot,**

In partial fulfilment
of the requirement for the degree of
Bachelor of Science in Information Technology,
Session (2022-2026).

By

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DECLARATION

We hereby declare that the project titled “Servista” is an original work carried out by us. This software and its accompanying documentation have been developed entirely based on our own efforts and ideas. No part of this project has been copied or taken from any existing source, software, or project.

If any part of this work is later found to be reproduced or copied from another source, we shall take full responsibility for the consequences. Furthermore, this project has not been submitted for any other degree or qualification at any other university or institute of learning.

Signature_____

Tehreem Aman

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Kashaf Mubarak

CERTIFICATE OF APPROVAL

It is to certify that the final year Project (FYP) of **BSIT**, session (2022-2026), titled “**Servista: An All-in-one Service providing platform**” was developed by **Tehreem Aman, Meerab Tasleem & Kashaf Mubarak** under the supervision of **Prof. Abu Zar Tamimi**.

In my opinion, this project after careful evaluation, the project has been found complete, well-developed, and suitable in both scope and quality for the degree of ***Bachelor of Science in Information Technology***. The project successfully presents an online platform where users can easily hire professionals for different home and technical services such as cleaning, plumbing, electrician work, tutoring, and maintenance.

FYP Advisor: Prof Abu Zar Tamimi Sb.

Advisor’s Signature: _____

Head of Department: Prof. Mussadiq Fida Sb.

HOD’s Signature: _____

EXECUTIVE SUMMARY

The main goal of Servista is to make it simple and quick for anyone to get reliable services without wasting time searching or calling multiple people. Through this platform, customers can book services online, check prices, and even review the service quality. On the other hand, service providers can register on the platform to offer their services and grow their business.

Servista is designed using modern web technologies to ensure it runs smoothly and securely. It includes features like user registration, service booking, admin control, and real-time communication between customers and workers.

Unlike traditional methods where finding help is time-consuming, Servista provides a digital and organized way to get things done making life easier for both customers and service providers.

In short, Servista is a smart, user-friendly, and trustworthy platform that brings all kinds of services to your fingertips, helping people save time, effort, and money through an easy online solution.

Keywords: [Servista](#), [web technologies](#), [admin control](#), [trustworthy](#).

ACKNOWLEDGEMENT

First, we thank Almighty Allah for giving us the strength and ability to complete our final year project, “**Servista: An All-in-One Service Providing Platform.**”

We are very thankful to our respected supervisor, ***Prof. Abu Zar Tamimi***, for his kind guidance, help, and throughout this project. His advice and encouragement made this work possible.

We also thank all our teachers for sharing their knowledge and helping us learn the skills needed for this project.

Special thanks to our friends and classmates for their ideas, motivation, and support.

Lastly, we are grateful to our families for their continuous love, prayers, and encouragement that kept us going until the end support.

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Tehreem Aman

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Kashaf Mubarak

ABBREVATIONS

BSIT	Bachelor of Science in Information Technology
FYPD	Final Year Project Documentation
UI	User Interface
SRS	Software Requirement Specification
URL	Uniform Resource Locator
HTTPS	Hypertext Transfer Protocol
ERD	Entity Relationship Diagram
API	Application Programming Interface

FYPD Overview

FYPD Title: Servista

Sr. No	Roll Numbers	Name	Signatures
1.	090467	Tehreem	
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Table 1 Project Proposal Summary

FYPD Goals
1. Develop a digital platform that connects users with reliable home service exports. 2. Simplify the process of Hiring, Electricians, Plumber, Cleaners, and home maintenance services. 3. Create a user-friendly website that offers fast, trusted and affordable services. 4. Create opportunities for service providers to reach a larger customer base and grow their businesses.
FYPD Objectives
1. Provide real-world services after natural disasters like flood, storm, earthquake, hailing. 2. Implement a fast-booking system and affordable and budget friendly environment. 3. Ensure positive feedback and customers satisfaction. 4. Build a responsive web application named Servista.
FYPD Success Criteria
1. This platform allow user to book and confirm a service successfully. 2. Experts can register, accept and complete service requests by doing so they can earn money. 3. Customer can view and give ratings and leave feedback after services. 4. System runs efficiently and effectively. 5. User can be able to chat with expert to discuss their specific requirements and needs.
Assumptions:
1. Experts are available in city and are verified by admin. 2. Payments and scheduling will be done online. 3. There is growing demand for online service platforms. 4. Users are increasingly comfortable with online booking and transaction.
Risks and Obstacles
1. Possible delays due to server or internet problems. 2. Inaccurate data entered by the user during registration or booking. 3. If the user didn't comply with rules, regulations or industry standards.
Organization Address: Government Jinnah Islamia College, Sialkot, Pakistan
Target End Users: Busy individuals seeking Hassle-free services.
Suggested Project Supervisor: Muhammad Abu Zar Tamimi
Approved By: Muhammad Abu Zar Tamimi
Date: 20 February 2026

Table 2 FYPD Goals

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CHAPTER NO 01: PROJECT PROPOSAL

1.1 Introduction:

Life gets busy, keeping up with household chores can be overwhelming, but Servista is here to make life easier. We bring together trusted experts for cleaning, plumbing, painting, stitching, solar maintenance, Pest control and many other home services in all-in-one places.

At Servista, we make it easy for you to get things done around the house without the stress. Whether you need help with cleaning, plumbing, painting, stitching, solar maintenance, or any other household task, we've got trusted professionals ready to help. We believe in quality work, fair prices, and friendly service. Just tell us what you need, and we'll take care of the rest so you can enjoy a clean, safe, and well-maintained home. Our platform is designed to be simple and user-friendly, so booking a service takes just a few clicks. You'll find detailed profiles, honest reviews, discounts and clear pricing no surprises, no confusion.

At Servista, we care about your comfort and peace of mind. That's why we only work with verified experts who treat your home like their own. Whether it's a quick fix or a big project, we're here to make your life easier. So, sit back, relax, and let Servista handle the chores because your time is valuable, and your home deserves the best.

1.2 Background:

We all know how stressful it can be to manage things at home whether it's a broken tap, a messy room, or a fan that just stopped working. Finding someone reliable to fix it can take time and effort. That's why **Servista** was created to make your life easier.

Servista is a place where you can find trusted help for all kinds of home services like cleaning, plumbing, painting, stitching, solar maintenance, and much more. Instead of searching here and there, you can simply visit Servista and book out the service you need in just a few clicks.

We also understand that sometimes life brings unexpected challenges like floods. After heavy rains or flooding, homes often need deep cleaning, water removal, sofa cleaning, wall painting, ceiling and sanitization to make them safe again. Servista provides reliable cleaning teams who can handle these tough situations, helping families get back to normal faster.

We know how important it is to feel safe and confident when letting someone into your home. That's why we only work with verified professionals who know their job and treat your home with care.

Our goal is simple to save your time, reduce your stress, and make sure your home stays in great shape whether it's everyday chores or emergency cleaning after a flood.

At Servista, we believe in using smart tools to make your life easier. Just like modern platforms use AI to match homeowners with the right service experts quickly and safely.

We're working toward a future where booking a plumber, cleaner, or solar technician is as simple as tapping a button. With secure communication, verified professionals, and clear service details, Servista helps you avoid confusion and get the job done right. Whether it's routine chores or urgent cleaning after a flood, our goal is to make sure your home gets the care it needs with trust, speed, and comfort.

At Servista, we're not just about fixing things we're about making life smoother through smart technology. As we grow, we're working on features like intelligent service matching, where the system learns your preferences and connects you with the best experts for your needs. We also plan to add real-time chat, secure payments, and easy tracking, so you always know what's happening with your service.

Our goal is to build a platform that's not only helpful today but keeps getting better tomorrow, so your home stays in good hands, no matter what.

1.3 Problem Statement:

People often struggle to find good and reliable help for everyday tasks like cleaning, plumbing, stitching, painting, and home repairs. Most of the time, they must visit service providers in person or call them manually, which takes a lot of time and causes delays. Even online platforms don't make things easy. They don't match customers with the right service providers based on location, availability, or skills. This leads to confusion, missed appointments, and poor service.

Another big issue is communication. Customers can't easily talk to service providers to ask questions or get updates. That's why our platform will include live chat, so people can connect instantly and get things done faster. Also, there's no proper way to check if a service provider is trustworthy. Without secure verification, customers may feel unsafe or unsure about who's coming to their home.

All these problems make the process stressful for both customers and service providers. It wastes time, causes frustration, and lowers trust. To solve this, we need a smart and easy-to-use platform that brings all services together, offers live chat, and ensures safety and making the whole experience smooth, quick, and reliable.

1.4 Stakeholders & Interests

Stakeholders	Their Roles
Homeowners & Users	Book services (cleaning, plumbing, painting, stitching, solar maintenance); view provider profiles, prices, and reviews; communicate and confirm bookings.
Service Providers	Create service listings, set availability and pricing, accept jobs, deliver services, and update job status.
Servista Team	Oversee operations, verify providers, handle user verification, prevent fraud, and ensure policy compliance.

Developers & Designers	Build and maintain the platform, improve performance, security, and reliability, and ship new features.
UI/UX designers	Make the interface simple, accessible, and easy to navigate; improve booking flow and clarity.
Marketing & Outreach Team	Promote Servista to households and communities, run campaigns, and grow both users and providers.
Customer Support Team	Resolve user issues, assist with bookings and payments, handle complaints, and collect feedback.

Table 3 Stakeholders & Interests

1.5 Objectives:

- Servista's main objective is to create a platform where users can access many services in one place.
- Servista aims to combine different services- such as home services, repair cleaning, maintenance, or professional help – into a single convenient system.
- The project focuses on giving users an easy process to search, select, and book any service without confusion.
- This platform ensures secure bookings, verified service providers, and clear communication between users and service workers.
- Servista helps skilled workers or service providers reach more customers and receive job requests easily.
- The system also keeps records of bookings, payments service history, and provide details to reduce manual work.
- User can review services, helping improve quality and helping others choose the best service providers.

1.6 Scope:

Servista covers a wide range of services such as home maintenance, cleaning, repairing services, technical support, and other daily need services. The system allows both customers and services providers to create accounts, manage their profiles, and update personal information. User can search for available services, view details (pricing, description, availability), and make bookings directly through the platform. Service provider can add, update, and manage the services they offer, including cost, schedule, and availability. Servista include options for online payments or cash on delivery, depending on the design of the system. The admin can manage users, service providers, service categories, bookings, complaints, and system reports. Users can view past booking, track current services status, and manage future service's needs. The platform includes filters like prices category, discount, offers/deals, location, or provider rating to help users find the right service quickly.

1.7 Assumptions:

The development, implementation and success of the servista website depend on the several critical assumptions that shape their functionality and user experience. These assumptions are

based on technological viability, user behaviour and industry standards for online home services provider platform.

- **Users have internet access:**
It is assumed that customers and service providers have a stable internet connection to access the servista platform.
- **Users have basic technical knowledge:**
The system assumes that user know how to use a smartphone or computer for tasks like creating accounts, booking services, and making payments.
- **Service providers are reliable and verified:**
It is assumed that all providers registered on the platform can deliver the services they claim and follow the platform rules.
- **Service information is accurate:**
The system assumes that the details added by service providers (price, timing, description) are correct and updated.
- **Users provide correct personal information:**
The system assumes that users enter true details during registration and booking.
- **Service availability depends on location:**
It is assumed that some services may only be available in certain areas, depending on where providers operate.
- **Admin manages the platform responsibly:**
It is assumed that the admin will regularly monitor and maintain the platform to ensure smooth operation.
- **Payment systems work properly & devices support the platform:**
It is assumed that online payment gateways used in servista are secure and functioning without issues. The website is assumed to work properly on commonly used devices like smartphones, tablets and computers.
- **Users follow platform rules:**
The system assume users will not misuse the platform, cancel booking without reason, or misuse reviews.

1.8 Risks:

In today era, where we enjoy comforts of technology it sometime also brings risk in your life. Like every technology which have some pros but also have cons/risk at the same time. **Servista** also have its pros and cons/risk. Some of its risks are as follow:

- **Security risk:**
Risk of having sensitive information of client stolen like username, home address account password since we handle booking and payment and many more.
- **Risk of fraud and misuse:**
Risk of false user profile, fraudulent application and deception after enjoying full services. The potential improper use of communication characteristics for spam or irrelevant interaction.
- **Project creep risk:**

Frequent additions or changes of attributes can increase the workload, extend development and shape capabilities.

- **Data handling risk:**
When the website collects customer information, the data must be stored, managed and protected properly. Any weakness here becomes a data management risk.
- **Server overload risk:**
Overloading of booking lead to the server crash, causing the website to crash or freezes.
- **Technical issue or delay:**
Technical issues like server overload, maintenance downtime and poor internet connection on the user's side can cause delay in work.
- **Performance risk:**
If a website is slow, customers will leave. The performance of the website is slow because of the factors like slow page loading, lengthy booking forms. If we want to build a good website it must be fast and reliable.

1.9 Success Criteria:

These are the must have features that will make Servista a trusted and easy-to-use platform for everyone. Below is the feature checklist that must be present for the acceptance of the Servista web application:

Category	Description
User Registration & Authentication	Secure sign-up/login with email and social media options, plus verification to keep accounts safe.
Service Listings & Smart Filters	Users can browse services like cleaning, plumbing, painting, stitching, solar, etc., using filters for location, price, availability, and urgency.
AI-Powered Service Matching	Smart recommendations based on user preferences, location, and service history.
Booking & Application Process	Easy-to-use booking forms, automated checks for service eligibility, and upload options for any required documents.
User & Provider Profiles	Both customers and service providers can create profiles with verification badges and service history.
Messaging System	Real-time chat between users and service providers with secure and private communication.
Post-Service Support	Follow-up options, maintenance tips, rebooking features, and access to trusted help resources.
Data Management & Storage	Secure database for user and service info, with regular backups and recovery options.
Search & Filter Options	Quick service search using filters like availability, ratings, specialties, and urgency.
Speed & Responsiveness	Fast-loading pages and smooth navigation for a great user experience.

Scalability	Platform can grow to handle more users, services, and bookings without slowing down.
Data Privacy & Security	Encrypted data, compliance with privacy laws, and protection against breaches.
Access Control	Role-based access for admins, customers, and service providers with permissions.
User Interface & Experience	Intuitive, user-friendly UI following accessibility standards ((WCAG (Web Content Accessibility Guidelines) compliance).
Cross-Platform Compatibility	Fully responsive design that works on desktops, tablets, and mobile devices.

Table 4 Success criteria

1.10 Tools, Libraries & Technologies:

Tools	Version	Rationale
VS Code	latest	Preferred for its lightweight and robust development environment.
Postman	latest	Used for API debugging and testing.
Git & GitHub	latest	Version control and collaborative development.
Docker	latest	Containerization for scalable deployment.
Jest	latest	Unit testing framework for JavaScript applications.

Table 5 Tools

Technology	Version	Rationale
React.js	latest	Ensure a responsive and interactive user interface.
Node.js	latest	Provides a scalable and efficient backend.
MongoDB	latest	NoSQL database for flexible and dynamic data.
AWS	latest	Cloud hosting and deployment for high availability.

Table 6 Technology

1.11 Work breakdown:

Sr. No	Roll no	Names	Team Roles & Responsibilities
1	090467	Tehreem	Front End, Back End & Documentation
2	090451	Meerab	Front End, Back End & Documentation
3	090486	Kashaf	Front End, Back End & Documentation

Table 7 Work division

1.12 Project planning:

	A	B	C	D
1	Task Name	Start Date	End Date	Duration(days)
2	planning	13-Oct-25	23-Oct-25	10
3	requirements gathering	25-Oct-25	15-Nov-25	20
4	analysis	16-Nov-25	29-Nov-25	13
5	designing	1-Dec-25	1-Jan-26	31
6	coding	13-Mar-26	25-Apr-26	42
7	testing	26-Apr-26	2-May-26	6
8	Implementation	3-May-26	15-May-26	12
9	Documentation	1-Dec-25	15-May-26	165
10				

Figure 1 project planning

1.13 Gantt Chart:

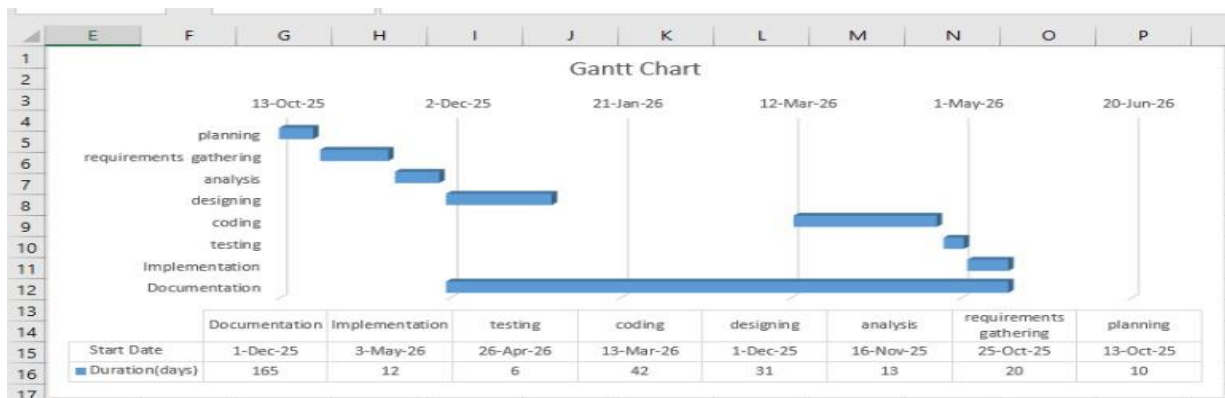


Figure 2 Gantt Chart

1.14 Conclusion:

Servista is built to make everyday life easier by bringing trusted home services like cleaning, plumbing, painting, stitching, solar maintenance, and more right to your doorstep. Our goal is simple to create a safe and user-friendly platform where people can quickly find the help they need without stress or confusion.

By combining smart technology with a human touch, Servista ensures smooth communication, clear service, and strong protections against fraud. We care deeply about privacy and make sure every user feels confident and comfortable using our platform.

As we grow, we'll keep listening to clients' feedback, improving our features, and exploring new ways to serve you better. Servista isn't just a service hub; it's your partner in keeping your home running smoothly every day.

We believe that everyone deserves easy access to quality help at home whether it's fixing a leak, refreshing a room with paint, or getting your solar panels checked. With Servista, you're not just booking a service you are joining a community built on trust, care, and convenience. Together, we're redefining what home support looks like: simple, safe, and always within reach.

CHAPTER NO 2 LITERATURE REVIEW

2.1 Literature Survey:

Servista is supported by a detailed study of modern service-providing platforms, home-maintenance apps, and real-time booking systems. Many existing platforms show how important it is to provide a smooth connection between customers and service providers, protect user data, and offer easy-to-use interfaces for booking home services. These studies help us design Servista as a secure, reliable, and user-friendly all-in-one services platform.

2.1.1 Digital platforms for home services:

Digital transformation has greatly improved how people book home services. Research shows that online service platforms increase service usage by providing quick access and verified professionals

Compared to traditional methods (like calling random workers), web applications give a faster, more comfortable, and more efficient way for users to find cleaners, plumbers, painters, tailors, solar technicians, and other service experts.

Pakistan based platforms (as a references)

► Kamkrew

<https://www.kaamkrew.com/>

Foreign websites

► Thumbtack (USA)

<https://www.thumbtack.com>

2.1.2 User experience and service booking rates:

Studies show that simple design, easy navigation, and personalized recommendations increase bookings on digital platforms.

Features like clear service categories, expert profiles, customer reviews, and interactive scheduling tools help users trust the platform and book services confidently.

When users can clearly see expert details and book instantly, overall satisfaction and platform engagement increase.

2.1.3 Challenges in online home-service platforms:

Although digital service platforms offer many benefits, they also face challenges such as fake service listings, unverified workers, hidden charges, and uncertainty about worker skills. Technical issues on the website, slow loading pages, errors in booking or payment failure reduce the user interest. During peak seasons (summer AC repairs, pre-Eid cleaning, monsoon plumbing issues) platform face too many bookings and failed to handle them efficiently. Users worry about their personal data, addresses and payment information being misused and leaked.

Solutions recommended by researchers include proper identity verification, detailed profiles of service providers, transparent pricing, and partnerships with trusted and verified technicians.

Servista follows these practices to ensure safety, trust, and high-quality service for every customer.

2.1.4 Ethical considerations and service reliability:

Ensuring ethical home-service practices is very important for Servista. Research shows that customers prefer platforms that send responsible, verified, and well-trained workers who respect the customer's home and privacy.

It is also important for digital platforms to ensure transparency, clear worker information, honest pricing, and proper communication so customers can make safe and confident decisions. Servista focuses on worker background checks, training, and accountability to maintain trust and high service standards.

2.2 Related Work:

Home-service platforms have upgraded from traditional methods of finding workers through phone calls or local shops to modern digital systems. These online platforms make it easier for people to book services like cleaning, plumbing, electrical repairs, painting, stitching, and solar maintenance.

Several existing apps and studies aim to improve user experience by introducing features such as smart matching, real-time communication, and secure payments.

Servista builds on these ideas by offering a complete, all-in-one solution that connects users with reliable service professionals quickly and safely, making home maintenance more convenient than ever.

Existing digital Service providing platforms:

✓ Kamkrew

Description	Kamkrew is a digital platform that connects customers with different home service providers. People can book services like cleaning, repairs, and maintenance through their app.
Features	• Online booking for various home services • Verified workers available in multiple cities • Price estimates before booking • Customer reviews and ratings
Limitations	• Limited-service categories compared to a full home-service platform • Availability depends on worker schedules • Some areas may not have fast response times

✓ Karsaz

Description	Karsaz is an online service marketplace that allows users to hire technicians such as electricians, plumbers, painters, and tailors.
Features	• Easy service booking through website/app • Skilled workers for basic home tasks • Tracking of service requests • Budget-friendly service options
Limitations	• Fewer multi-service options compared to all-in-one platforms • Quality of service can vary depending on the individual worker • Not available in every city

✓ Service market.pk

Description	ServiceMarket.pk provides different home and office services, helping customers book cleaners, plumbers, electricians, and other technicians online.
Features	• Wide list of service categories • Transparent pricing for many services • Customer support for booking and complaints • Allows quick scheduling
Limitations	• Some services may not cover specialized home chores • Service prices can be higher than local market • Limited advanced services like stitching or solar solutions

Table 8 References

2.2.1. The importance of digital platforms in booking home services (Traditional Service Booking vs Digital)

Traditionally, people book home services through local shops, personal references, or printed ads. While these methods work, they come up with challenges such as limited availability, slow response times, and difficulty comparing service providers.

Digital service platforms like Servista solve these problems by:

- Expand access to more service providers beyond your neighborhood.
- Offering detailed service provider profiles with reviews, experience, pricing, and photos of past work, making it easier to choose the right professional.

- Improving decision-making with transparent ratings and verified customer feedback.
- Reducing communication barriers through in-app messaging and instant booking options.
- Saving time by allowing users to compare, schedule, and pay online without visiting multiple shops.

Servista is built on these digital strengths to make home maintenance easier, faster, and more reliable for every household.

2.2.2. User experience and participation in digital home service platforms

- Easy-to-use design and accessibility
- A smooth and user-friendly interface plays a big role in helping customers confidently book services. Research on digital service platforms shows that:
- Clear navigation and smart search filters encourage more users to explore and book services.
- Mobile accessibility is essential, as most people now book services directly from their phones.

Servista applies these insights by:

- Providing advanced filters based on service type, price range, location, availability, and customer ratings.
- Offering a responsive design that works perfectly on phones, tablets, and computers.
- Ensuring that users can quickly book trusted professionals with just a few taps.
- The role of high-quality images in service selection

How good images help customers choose services?

- Clear and high-quality photos or short videos of your service work (like before/after cleaning, plumbing fixes, painting finish, stitching results, solar setup, etc.) help customers trust your service.
- Customers prefer service providers who show neat, real, and professional-looking examples of their work.
- Using AI-enhanced images can make your service listings look clean and more appealing.

Servista improves customer choice by:

- Allowing service providers to upload multiple photos and short video clips of their work.
- Using AI tools to automatically improve the clarity and quality of uploaded service photos.

2.2.3. The role of technology in improving service booking efficiency:

- Online booking and automated processing
- Traditional service booking (calling, visiting, waiting for confirmation) often takes time.
- Online booking makes the process faster and easier.
- Technology helps speed up service booking by reducing manual work and waiting time.
- Providing proper forms so customers can clearly explain the service they need.

- Matching customers with the right service provider quickly.
- *Servista* improves booking efficiency through a structured online booking system where service requests can be approved, assigned, or rescheduled easily.
- Automated notifications that update customers about the status of their service request (confirmed, on the way, completed, etc.).
- Communication and transparency for service users
- Studies show that clear communication between customers and service providers reduces confusion and improves satisfaction.
- *Servista* supports better communication by allowing in-app messaging so customers and service providers can discuss details directly.
- Providing AI-powered help (chatbot) to answer common questions instantly.
- Making the whole process transparent customers can track who is coming, when they are coming, and what the service includes.

2.2.4. The impact of location-based services on booking success:

How location helps customers and service providers?

- Location-based services make it easy to connect customers with the nearest available service provider.
- This increases the chances of getting faster service and reduces waiting times.
- It also decreases travel time and cost for service providers.
- *Servista* uses location benefits by showing nearby professionals who can quickly reach the customer.
- Reducing travel costs and delays for both customers and service providers.
- Making the overall service experience smoother and more efficient.

2.2.5. The role of digital platforms in improving household service accessibility and sustainability:

- Digital service booking and support for service providers
- Many local service providers struggle with low visibility, inconsistent work, and lack of proper promotion.
- Digital platforms like *Servista* help solve these challenges by reducing workload gaps for service providers by increasing their visibility to customers.
- Encouraging professional and responsible work practices through ratings and feedback.
- Providing financial stability by connecting skilled workers with more regular job opportunities.
- Helping customers make informed choices by offering transparent profiles, pricing, and reviews.
- *Servista* supports service providers by offering tools to manage bookings, track earnings, and maintain service history.
- Providing educational resources to improve their skills and service quality.
- Creating opportunities for additional income and professional growth.

2.2.6. Safety and ethical considerations in using digital home service platforms

Data Safety and Privacy

- As digital bookings become more common, data privacy becomes extremely important. Users must feel safe while sharing personal details and booking services online.
- *Servista* ensures this through end-to-end encryption to protect user data and communication.
- Multi-factor authentication (MFA) for safer account access.
- Strict compliance with data protection laws and privacy standards.

Ethical Service Practices

- Digital service platforms must prevent harmful or unethical behavior such as fake service providers, fraud, overcharging, or poor-quality work.
- *Servista* promotes ethical service through strict verification of service professionals to ensure reliability.
- Background checks when necessary (e.g., for in-home services).
- AI-powered fraud detection to identify suspicious behavior.
- A transparent rating and review system to maintain quality and trust.

2.3 Gap Analysis:

Despite the availability of many home service platforms, most still have gaps that affect user experience, trust, and efficiency. These gaps include:

- Lack of verified and skilled professionals
- Poor communication systems
- No proper customer supports
- Unclear pricing and hidden charges
- Limited-service categories

1. Lack of smart service recommendations

Many existing platforms rely on manual browsing, which is slow and confusing. Users struggle to find the right service provider for their exact needs.

2. Weak real-time communication

Most platforms do not allow quick messaging or live updates between customers and service professionals. This leads to delays, misunderstandings, and frustration.

3. Poor verification of service professionals

Fake listings and unverified workers can create safety risks. Without proper identity checks, users cannot trust who is coming to their home.

4. Limited support after booking

Current platforms often fail to provide follow-up assistance, such as issue reporting, service warranties, or customer support after work is completed.

5. Bad user experience (UX)

Many platforms have old designs, slow loading times, complicated booking steps, and limited accessibility, making the whole booking process difficult.

2.4. Summary:

The literature review shows how digital platforms have changed the way people book home services by making the process faster, more accessible, and more convenient. It highlights the importance of user-friendly design, smart service matching, real-time communication, and verified service professionals in building trust and improving customer satisfaction. However, many existing platforms still struggle with major issues such as weak verification systems, lack of AI-based recommendations, poor user experience, slow communication, and limited support after a service is completed. To address these problems, *Servista* proposes a modern, reliable, and easy-to-use platform where customers can quickly find trusted professionals for cleaning, plumbing, painting, stitching, solar services, and other home maintenance tasks. By offering transparent

With pricing, secure communication, strong verification, and improved service management tools, *Servista* aims to enhance user experience, simplify the booking process, and build long-term customer confidence.

CHAPTER NO. 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Requirement Analysis:

The software Requirements Specification (SRS) for the Servista web application describes the functional, non-functional, user, and system requirements of Servista, a web-based platform that connects customer with service providers offering a wide range of services such as home maintenance, cleaning, painting, repairs and professional support. This analysis ensures that the system satisfies the needs of different user, including customers, service providers, and administrators. It details the objectives, scope, interested parties, functional module system architecture, limitations and security considerations. The document serves as a plan for developers, ensuring that the system meets the user's needs while maintaining scalability, performance and data safety. Through well-defined requirements, the SRS guarantees that servista delivers a safe, easy to use and safe services experience.

3.2 User classes and characteristics:

1. Customers (Services Seekers)

Description: Individuals who use the website to book home or personal services.

User class	User Characteristics
Customers	Users who book services and need a simple, fast, and reliable platform to find trusted service providers.

2. Service Providers (Technicians/Vendors)

Description: Professionals offering services through the platform (e.g., plumbers, electricians, cleaners, beauticians)

User class	User characteristics
Service Providers	Skilled professionals who use the platform to receive job requests, manage schedules, and earn income.

3. Administrators (Platform Managers)

Description: Individuals managing the servista platform backend.

User class	User characteristics
Administrators	Staff responsible for managing users, monitoring service quality, resolving issues, and maintaining system operations.

Table 9 User classes and characteristics

3.3 Requirement Identifying Technique

In identifying the functional requirements for Servista, the use case technique is applied because servista is an interactive web-based home service platform. This technique allows defining interactions among customers, service, providers and administrators for efficient system operation.

The system is designed at another level of detail for each service function. Every functionality is simplified in a way that captures the main user actions with the expected system

actions in return. These interrelations are captured in separate use case diagrams and are described in full dressed use cases.

These are some of the most important use cases and their corresponding requirements:

Record Use Case	Associated Use Case Requirements
User Registration	Users must be able to register by providing personal details such as a name, phone number, and address.
Service Booking	Customers should be able to book a service from available categories (plumbing, electrical, and cleaning, etc.).
Service Provider availability	Service providers should be able to update their availability and working schedule in the system.
Service Tracking	Customers should be able to track the status of their booked service in real time.
Payment processing	The system must allow users to make secure payments via online gateways or cash-on-delivery options.
Service history	Users should have access to their previous services records for reference or rebooking.
Provider Assignment	The system should automatically assign available and nearby service providers to customers.
Rating and Reviews	Customers should be able to submit feedback after service completion.

Table 10 Record use case

3.4 Functional Requirements

Functional requirements define how the servista system should operate and the actions it must support for customers, service providers, and administrators.

3.4.1.FR-1: User Registration and authentication

Identifier	FR-1
Title	User Registration and authentication
Requirement	The customer, service provider, or administrator shall be able to create an account, log in, and manage authentication securely using an email /phone number and password. Service providers shall also be able to upload identification documents for verification.
Source	System requirement
Rationale	Required for secure access control, user identity verification, and personalized services experience on the servista platform.
Business Rule	User credentials must be encrypted and comply with security standards. Duplicate emails/phone numbers shall not be allowed. Servista providers must upload valid documents for verification.
Dependencies	Required for service browsing and search (FR-2), service booking (FR-3), and provider assignment (FR-4).

Priority	High
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Table 11 User Registration and authentication

3.4.2 Service browsing and search

Identifier	FR-2
Title	Service browsing and search
Requirement	The customer shall be able to view available service categories (plumbing, cleaning, electrical, and solar installing etc.) and search for specific services based on location, category, or keywords.
Source	User requirement
Rationale	Enable customers to quickly find appropriate services and make informed decisions.
Dependencies	Requires user profile (FR-1) and service database setup.
Priority	High

Table 12 Browsing and search

3.4.3 Service Booking

Identifier	FR-3
Title	Service Booking
Requirement	The customer shall be able to book a service by selecting a provider, data, time, and service location. The system shall confirm availability before completing the booking.
Source	Core user requirement for service request and scheduling.
Rationale	Provides the main purpose of the platform –scheduling home services.
Business rule	Booking may not be created without verified user information. Double booking the same provider at the same time is not allowed.
Dependencies	Depends on provider availability (FR-4) and user registration (FR-1).
Priority	High

Table 13 Service booking

3.4.4 Provider Availability Management

Identifier	FR-4
Title	Provider Availability Management
Requirement	Service providers shall be able to update their availability (dates, time slots) so that the system can assign them to customer bookings accordingly.

Source	Operational requirement for coordinating service assignments.
Rationale	Ensures accurate scheduling and prevents conflicts or double bookings.
Business rule	Providers must maintain up-to-date availability to remain eligible for booking.
Dependencies	Required for services booking (FR-3) and provider assignment processes.
Priority	Medium -High

Table 14 Provider Availability Management

3.4.5 Real time Service Tracking

Identifier	FR-5
Title	Real time Service Tracking
Requirement	The customer shall be able to track the status of their booked service in real-time (e.g., Accepted, on the way, in progress, completed). Providers shall update their status from their interface.
source	User requirement for transparency and service monitoring.
Rationale	Enhances customer trust and provides visibility into the progress of the service request.
Business rule	Status updates must be accurate; providers must update status at each stage.
Dependencies	Depends on service booking (FR-3) and provider profiles (FR-1).
Priority	Medium

Table 15 Service tracking

3.4.6 Secure Payment Processing

Identifier	FR-6
Title	Secure Payment Processing
Requirement	The system shall allow customers to make payments using multiple methods such as credit/debit cards, mobile wallets, or cash on delivery. The system shall validate and confirm payment before booking completion.
Source	Business requirement for payment verification and service charges.
Rationale	Enables revenue processing and ensures secure financial transactions.
Business rule	Payments must use encrypted channels; unauthorized transactions must be flagged or rejected.
Dependencies	Requires service booking (FR-3) and user registration (FR-1).
Priority	High

Table 16 Secure payment processing

3.4.7 Ratings and Reviews System

Identifier	FR-7
Title	Ratings and Reviews System

Requirement	The customer shall be able to rate service providers and write reviews after a service is completed. The system shall store this feedback and display average ratings on provider profiles.
Source	User experience requirement to evaluate service quality.
Rationale	Helps maintain service quality and assists users in choosing reliable providers.
Business rule	Reviews can only be submitted after service completion; inappropriate content may be removed by admin.
Dependencies	Depends on completed service status (FR-5).
Priority	Medium

Table 17 Ratings and Review system

3.4.8 Admin Management Dashboard

Identifier	FR-8
Title	Admin Management Dashboard
Requirement	The administrator shall be able to manage users, service providers, service categories, complaints, and system data using an admin dashboard.
Source	Business requirement for system maintenance and oversight.
Rationale	Ensures the smooth operation of the platform and maintains data integrity.
Business rule	Only authorized admins can access the dashboard; admin actions must be logged for security.
Dependencies	Depends on user registration (FR-1), service categories, and booking data.
Priority	High

Table 18 Admin management dashboard

3.4.9 Notification system

Identifier	FR-9
Title	Notification System
Requirement	The system shall send notifications to customers and service providers regarding booking confirmations, cancellations, service status updates, and payment confirmations.
Source	User requirement for timely updates and communications
Rationale	Ensures users are informed about important events, improving service coordination.
Business rule	Notifications must be triggered in real time and must match system events accurately.
Dependencies	Depends on service booking (FR-3), payment processing (FR-6), and service tracking (FR-5).
Priority	Medium -High

Table 19 Notification system

3.4.10 Service History Management

Identifier	FR-10
Title	Service History Management
Requirement	The system shall store and display past service records both customers and service providers, including completed bookings, payments, and reviews.
Source	User requirement for record -keeping and reference.
Rationale	Helps users review past services and assists providers with performance tracking.
Business rule	Records must not be deleted unless required by legal or data policy rules.
Dependencies	Depends on service completion updates (FR-5) and payment records (FR-6).
Priority	Medium

Table 20 Service history

3.5 Non-functional Requirements

Non-functional requirements define how the system should behave and specify the performance, security, reliability, usability, scalability, and other quality standards a system must meet.

They do not define system features; instead, they define the constraints and behavioural qualities of those features.

3.5.1 Performance Requirements

- The system should load any page within 2-3 seconds.
- Response time should not exceed 5 seconds during high traffic.

3.5.2 Security Requirements

- User passwords must be encrypted.
- Only authorized users can access admin functions.
- The system must comply with data protection laws.

3.5.3 Usability Requirements

- The interface must be simple and easy to navigate.
- Users should be able to complete registration in under 2 minutes.

3.5.4 Reliability Requirements

- The system should have 99.5% uptime.
- The platform must recover automatically from minor errors.

3.5.5 Scalability Requirements

- The system must handle increasing users without performance degradation.
- Support up to 50,000 concurrent users.

3.5.6 Maintainability Requirements

- The system should allow developers to update modules without downtime.
- Code must follow standard naming conventions.

3.5.7 Availability Requirements

- The service should be available 24/7.
- Downtime for maintenance should not exceed 2 hours per month.

3.5.8 Compatibility Requirements

- Works on all major browsers (Chrome, Safari, Firefox, Edge).
- Supports web + mobile responsiveness.

3.6 External Interface Requirements

The servista web application must interact without problem with users, external software, and hardware systems. This section defines the user interface, software interface, hardware interface, and necessary communication protocols for smooth operation and integration.

3.6.1 User Interface Requirements

The system must provide a responsive, intuitive, and user-friendly interface to enhance the overall user experience.

Web based UI	• Build using React/Next.js with tailwind CSS to support a clean and responsive layout. • Optimized for desktop, table, and mobile screens for consistent performance across devices. • Supports dark mode, high-contrast mode, and large text settings for improved accessibility. • Ensures key actions (such as booking a service or viewing providers) can be completed within three clicks.
Interactive Elements	• Allows users to filter services based on category, location, provider rating, availability, and price range. • Displays service provider details such as skills, experience, reviews, pricing, and availability calendar. • Provides an interactive service booking from with selectable dates, times, and payment options. • Offers real-time chat between customers and service providers using a clean, WhatsApp-style chat box.

Table 21 User interface requirements

3.6.2 Software Interface Requirements

The system must interact with API and third-party services for improved functionality.

Payment Gateway	Integrates with secure payment systems for processing online payments.
Email/SMS API	Uses external services to send OTPs, booking confirmations, and updates.
Maps and location API	Integrates Google Maps API for location detection, route assistance, and service radius calculation.
Authentication Services	Supports OAuth/JWT for secure login, including optional social media login integration.
Database Interface	Connects with SQL/NoSQL databases to store user, provider, and booking information.

Table 22 Software interface requirement

3.6.3 Hardware Interface Requirements

Hardware interfaces although mainly a web-based platform, the system must admit external devices when necessary.

User Devices	Must operate on any internet enabled device including smartphones, tablets and desktops.
GPS-Enabled Devices	For location-based services, devices with GPS support help determine user or provider location.

Table 23 Hardware interface requirement

3.6.4 Communication Interfaces

The platform must admit the transmission of safe and efficient data between users and external systems.

HTTPS Protocol	All communication between client and server must use encrypted HTTPS connections.
REST APIs	The frontend communicates with the backend through secure RESTful APIs.
Real-time Notifications	Uses Web Sockets or push notifications for booking updates and in app messages.
Email /SMS Communication	Supports automated communication for reminders, OTP, and service updates.

Table 24 Communication interface

3.7 Use case Analysis

3.7.1 Use Case # 1: User Registration and Authentication

UC Identifier	UC1: Register and Authenticate User
Requirement Traceability	FR-1(User Registration and Authentication)
Purpose	To allow customers, service providers, and administrators to securely register and log in to the servista platform

Priority	High
Preconditions	The user has access to a valid email /phone number and the Servista login/registration interface.
Postconditions	The user is successfully authenticated and redirected to their respective dashboard (customer, provider, admin).
Actors	Customer, Service Provider, Administrator
Extends	None
Main Success Scenario	1. User enters registration details. 2. System validates inputs. 3. User receives OTP /email verification. 4. User log in 5. System grants access to the dashboard.

Table 25 User case 1

3.7.2 Use Case #2: Search and Filter Services

UC Identifier	UC2: Search and Filter Services
Requirements Traceability	FR-2(Service Browsing and Filtering)
Purpose	Allows customers to search for home services using filters such as category, price, availability, and location.
Priority	High
Preconditions	User is logged in or browsing as a guest; services are available in the system.
Postconditions	A list of services matching the user's filters is displayed.
Actors	Customer
Extends	Browsing Services
Main Success Scenario	1. User opens search page 2. Selects filters (e.g., category, location, price) 3. System retrieves matching services 4. User views services list.

Table 26 User case 2:

3.7.3 Use Case #3: Book a Service

UC Identifier	UC3: Book a Services
Requirements Traceability	FR –3(Service booking)
Purpose	Allows customers to schedule and book a service with a selected service provider.
Priority	High
Preconditions	User is logged in; service provider has an available time slot.
Postconditions	Booking is saved, and confirmation is sent to both customer and provider.

Actors	Customer, Service provider
Extends	Search and Filter Services
Main Success Scenario	1. User selects provider. 2. chooses date/time. 3. Enters address. 4. Confirms. 5. Payment completed. 6. Booking confirmed.

Table 27 User case 3

3.7.4 Use Case #4: Manage Service Availability

UC Identifier	UC4: Manage Services Availability
Requirements Traceability	FR –4(Provider Availability Management)
Purpose	Allows service providers to update their working hours, available dates, and prices.
Priority	Medium
Preconditions	Provider is logged into their account.
Postconditions	Updated availability is reflected in customer search results.
Actors	Service provider
Extends	None
Main Success Scenario	1. provider opens 2.availability page. 3. Edits schedule. 4. Saves changes. 5. System updates. 6. Availability calendar.

Table 28 User case 4

3.7.5 Use Case #5: Payment and Confirmation

UC Identifier	UC5: Payment and Confirmation
Requirements Traceability	FR –5(Payment processing)
Purpose	Allows customers to pay securely and receive booking confirmation.
Priority	High
Preconditions	User has selected a service and confirmed booking details.
Postconditions	Payment receipt generated; booking marked as confirmed.
Actors	Customer, Payment gateway
Extends	Book a Service
Main Success Scenario	1. User selects payment method. 2.System processes payment. 3. Payment succeeds. 4. Confirmation sent.

Table 29 User case 5

3.7.6 Use case Diagram

- **Customer Use Case Diagram:**

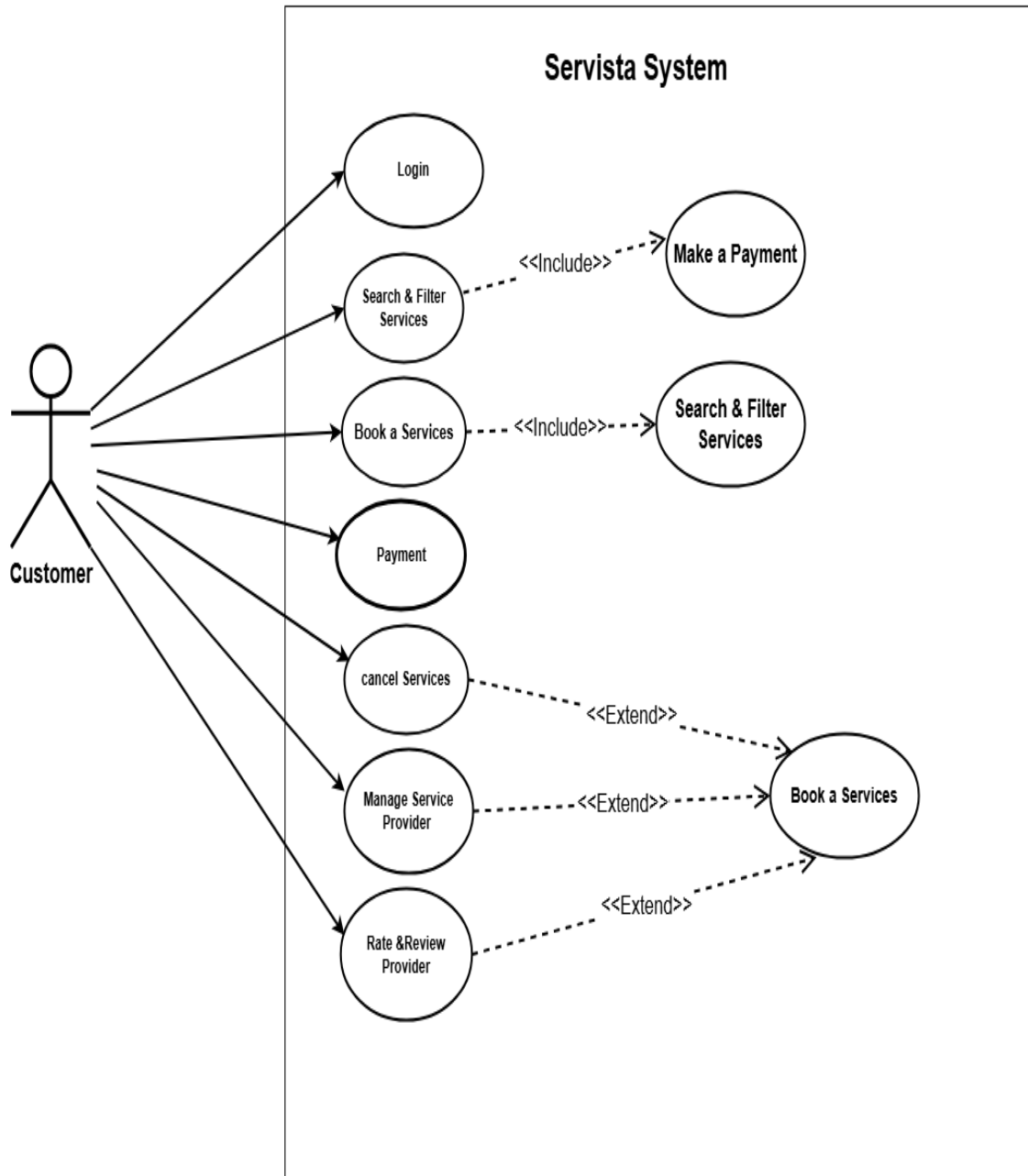


Figure 3 customer use case diagram

- **Service Provider Use Case Diagram:**

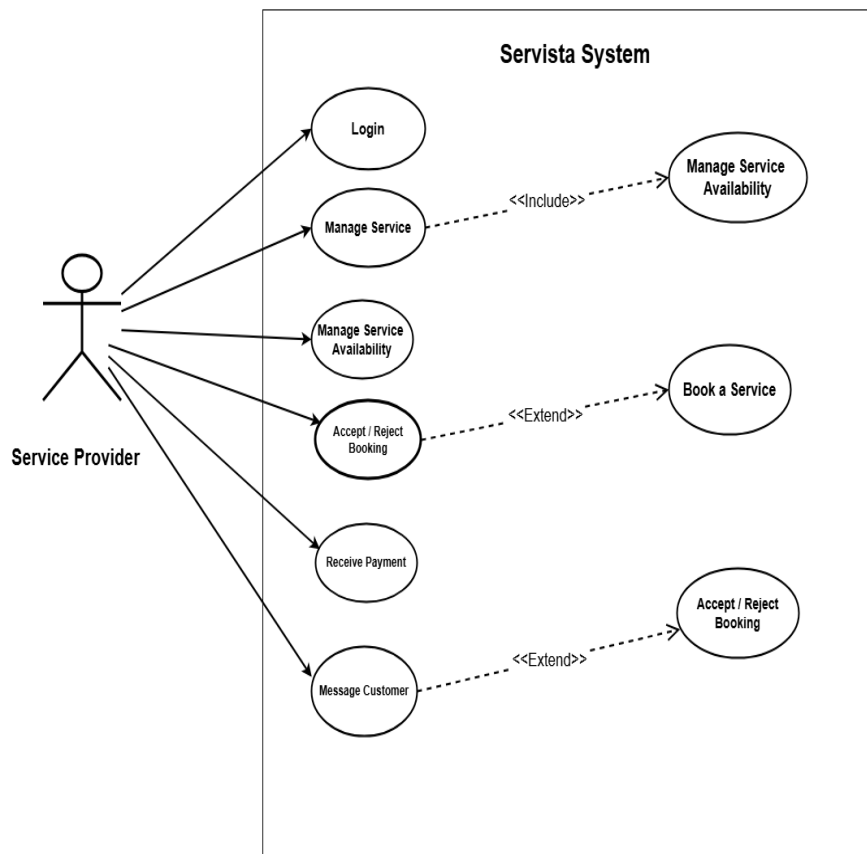


Figure 4 service provider use case diagram

- Admin Use Case Diagram:**

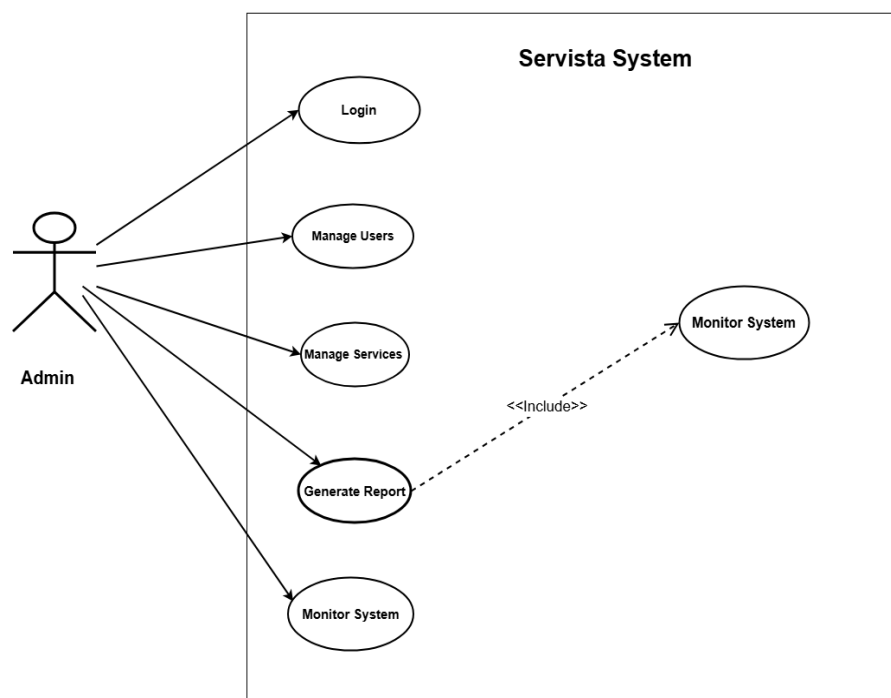


Figure 5 admin user case diagram

3.8 Storyboarding

Since storyboards visually represent user interactions with the system, I will describe three essential processes other than login or register and suggest creating corresponding diagrams for them.

3.8.1 Booking a Home Service

User Steps:

1. The customer select a service category (e.g., electrician, plumber, cleaner home repairing and maintenance, solar installation) from the list.
2. Choose a service provider and views available date and time slots.
3. Confirm the booking by providing address and additional notes.
4. Completes the payment (if required) and submits the request.
5. Receives a confirmation message via email/SMS or in-app notification.

System Response:

- The system checks the provider's availability for the selected time slot.
- Saves the booking details in the database.
- Sends a booking confirmation notification to both the customer and the service provider.
- Updates the provider's schedule and marks the slot as occupied.

Storyboard Diagram for booking Services

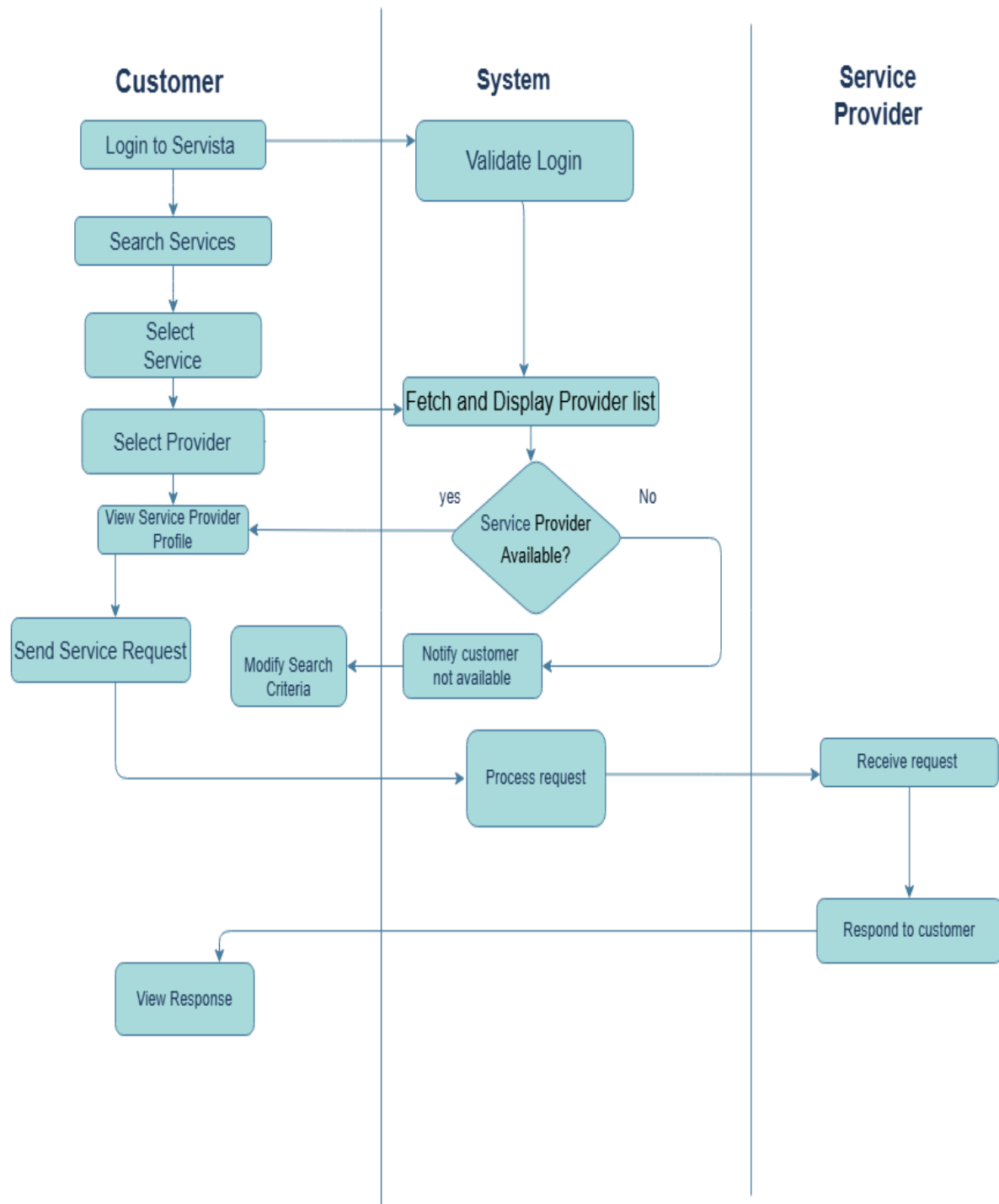


Figure 6 storyboard diagram for booking service

3.8.2 Service Provider Accepting a job

User Steps:

1. The service provider logs into their dashboard.
2. Views new service requests pending approval.
3. Reviews customer details, service type, and schedule time.
4. Accepts or declines the job request.

System Response:

- Displays all pending jobs in the provider dashboard.
- Updates job status to “Accepted” or “Declined” based on provider action.
- Notifies the customer instantly about the provider’s decision.
- Updates the provider’s availability calendar.

3.8.3 Completing a service and receiving a Review**User steps:**

1. The provider marks the service as “completed” after finishing the job.
2. The customer receives a notification to review the service.
3. The customers rate the service provider and writers a review.
4. Submits the feedback.

System Response:

- Updates the booking status to “completed”.
- Triggers a prompt for the customer to provide a rating and review.
- Saves the review and rating in the database.
- Updates the provider’s overall rating on their public profile.

3.9 Summary

The process of requirements identification and use case analysis plays a fundamental role in ensuring that the Servista system is designed and developed in alignment with user expectations, system objectives, and operational needs.

Requirements identification begins with gathering detailed information about stakeholder needs, including customer seeking home services, service providers offering that service, and administrators managing the platform.

Techniques such as interview, observations, use case modelling, and analysis of similar service platforms help identify both functional and non-functional requirements. These requirements define what the system must do, how it should perform, and the constraints it must be operate within.

Once the requirements are gathered, use case analysis provides a structure method to translate these needs into actionable system interactions.

Use cases clearly outline how different users interact with the system, the goals they want to achieve, and the steps necessary to reach those goals. Each use case specifies actors, preconditions, main actions and expected outcomes, ensuring that all system functionalities are well – understood before development begins. Through this analysis, developers can detect

dependencies, identify potential challenges and ensure that critical features- such as service booking provider management, and user authentication- are fully captured.

Overall, the requirements engineering process ensures that servista meets its project objectives by establishing a clear foundation for system design. It minimizes ambiguity, reduces development risks, and helps maintain alignment between stakeholder expectations and system capabilities. By thoroughly analysing requirement and modelling user interactions through use cases, the project gains a structured roadmap that guides development, enhance system usability, and supports the creation of a reliable, efficient, and user –centred home service management platform.

CHAPTER NO 4:

SOFTWARE DESIGN SPECIFICATION

4.1 System Design

The Servista Web Application is designed as a centralized digital platform that connects customers with trusted professionals for tasks like repairing, electrical work, carpentry, cleaning, and many more. The system work as an application based on the web that interact with multiple external services and API's which allow perfect integration with location services, real time booking notifications, secure payment gateway and vender verification system, enhancing the overall user experience.

The application follows a modular architecture, ensuring flexibility, maintenance ease, and more. Following are the central component of this Web Application:

- **Frontend (User Interface):**
Developed with React.js and CSS, to provide the responsive user interface for Customers, Service Providers, and Administrators.
- **Backend (Server Layer):**
Built with Python, managing Authentication, Service listing, Job requests and scheduling, and In-app messaging.
- **Database Management:**
Use MySQL for structured data storage (user profiles, services record, job orders, and customers reviews).
- **Third party Integrations:**
 1. Payment Gateways-secure transaction between customer bank card to your service provider.
 2. Service scheduling platform-it prevents double booking and matches job with skilled professionals.
 3. Customer Engagement Platform store and manage booking history, customer data and communication records in one place.

4.2 Design Considerations:

1. Assumptions and Dependencies:

Assumptions	• Trusted service hub • Seamless online booking and payment • Growing demand for home services online • Rising trust in online payment
Dependencies	• Reliable professional • Communication dependency • Financial dependency

2. Limitations:

Performance	• Database overload • Scalability bottleneck • Real time update
Functional	• Scheduling limitations • Service package limitations • Location based services • Payment process limitation
Regulatory and Compliance	• Data Privacy Constraints • Third-Party Service Dependency • Labors laws and regulations • Environmental regulations

Table 30 Design Consideration

4.3 Requirements Traceability Matrix:

Requirement ID	Scope	Requirement Description	Design Specification
R1	User Management	The system shall allow users to register, log in, and manage profiles securely.	Component “User Authentication & Role Management”
R2	Home service dictionary	The system shall allow users to search for services	component “service finder& Filtering Module”
R3	Service package management	Service provider shall be able to upload, update and manage services packages and discount in detail and with images.	component “service package management and database storage”
R4	Communication system	System shall allow real time massaging between customer and service provider	Component “real time messaging system”
R5	support	Provide user with support for booked services	Support and resources

R6	Scheduling	Give full control to service provider to manage their schedule and appointment	Scheduling module
R7	Event management	Allow service provider to promote their events and offers	Event notification
R8	Reporting and analytic	Generate report on bookings, payments and user activity	Reporting analytic dashboard
R9	Location based service	Help user find nearby service providers	Google map APIs integration

Table 31 requirement traceability matrix

4.4 Design Models:

Servista web application follows an object-oriented development approach, which incorporates several design models to illustrate the structure, behavior, and system interactions. The key models used to describe the system design are presented below: **4.4.1 Design Class Diagram (DCD):**

- **UML Class Diagram**

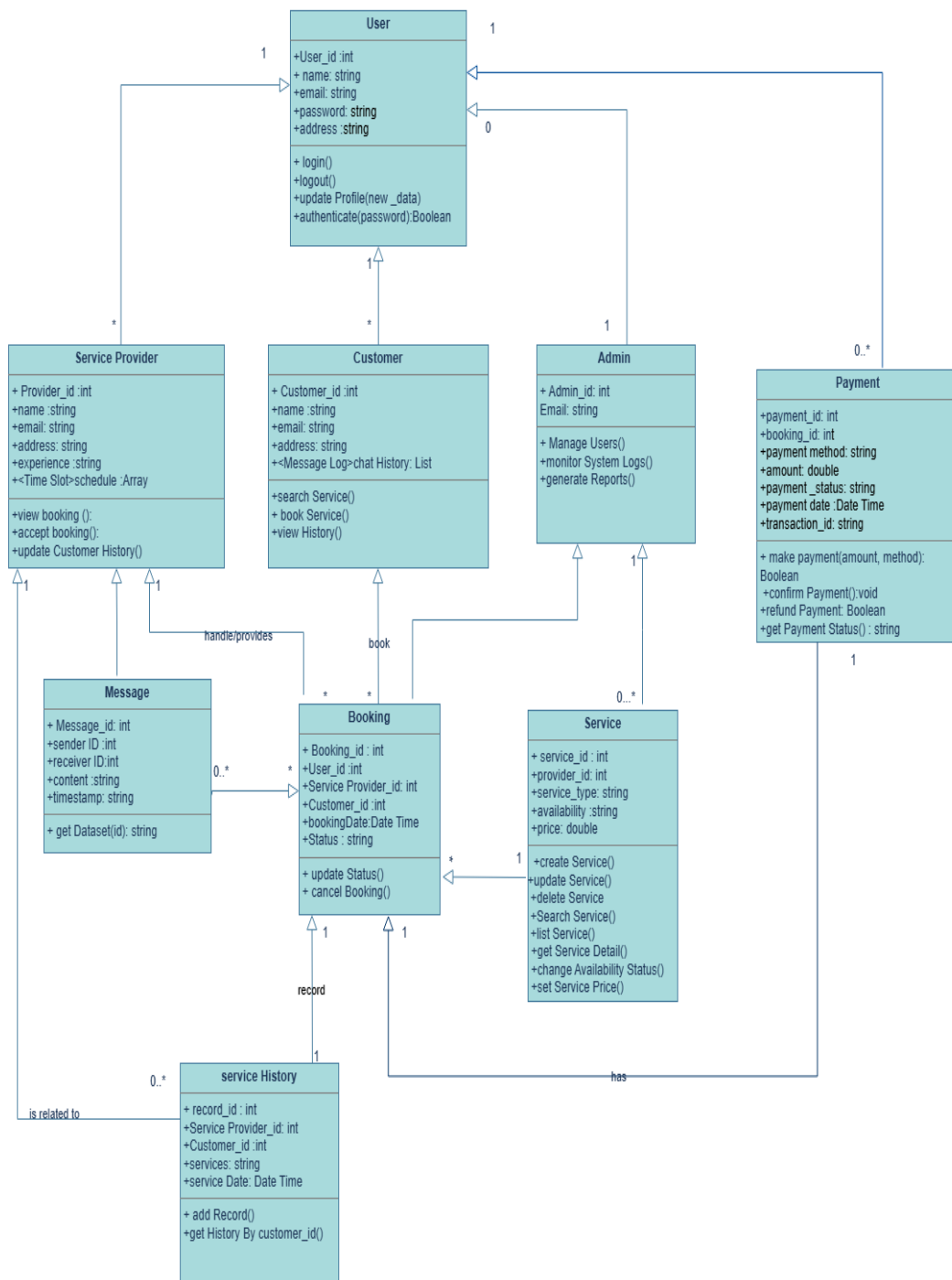


Figure 7 UML class diagram

4.4.2. Sequence Diagram:

1. User Authentication Sequence Diagram

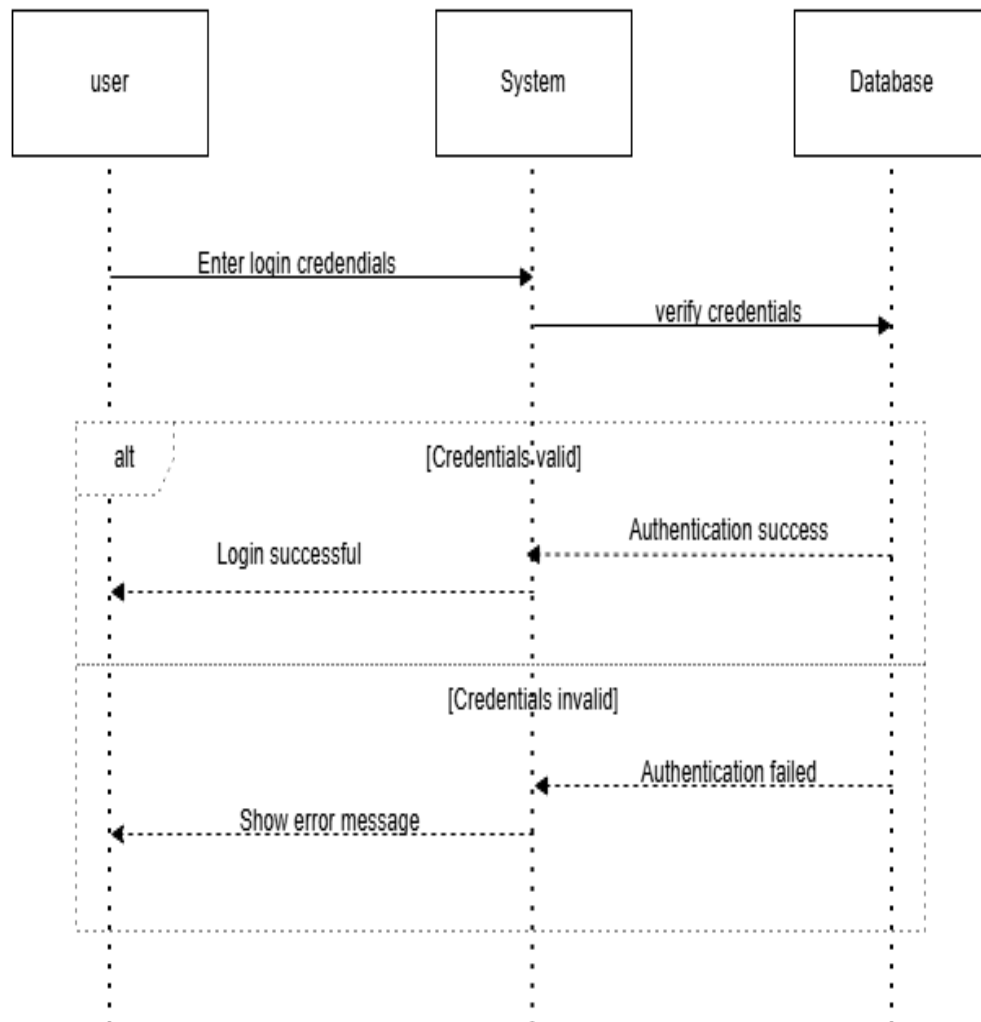


Figure 8 user authentication diagram

2. Service Search Sequence Diagram:

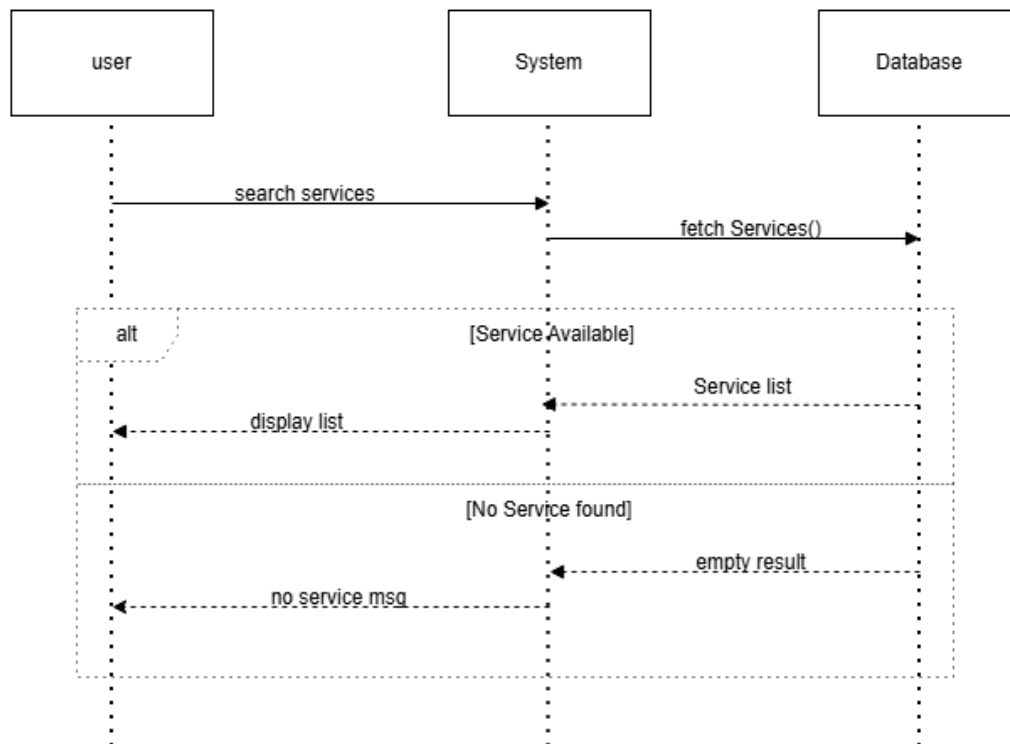


Figure 9 Service Search Sequence Diagram

3. Service Booking Sequence Diagram:

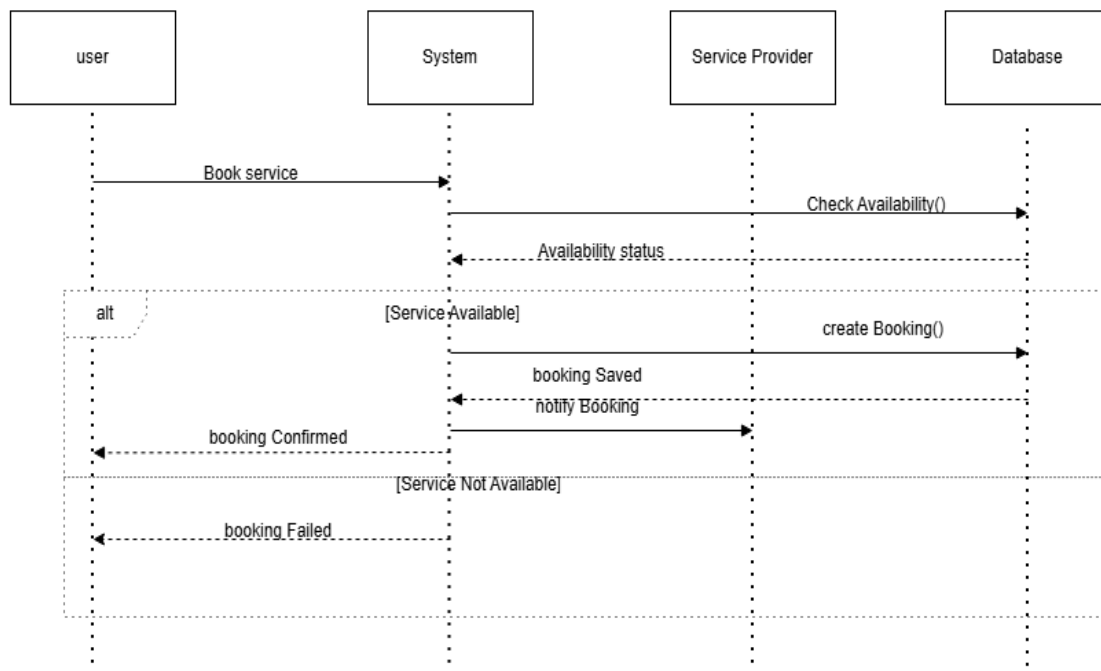


Figure 10 Service Booking Sequence Diagram

4. Payment Processing Sequence Diagram:

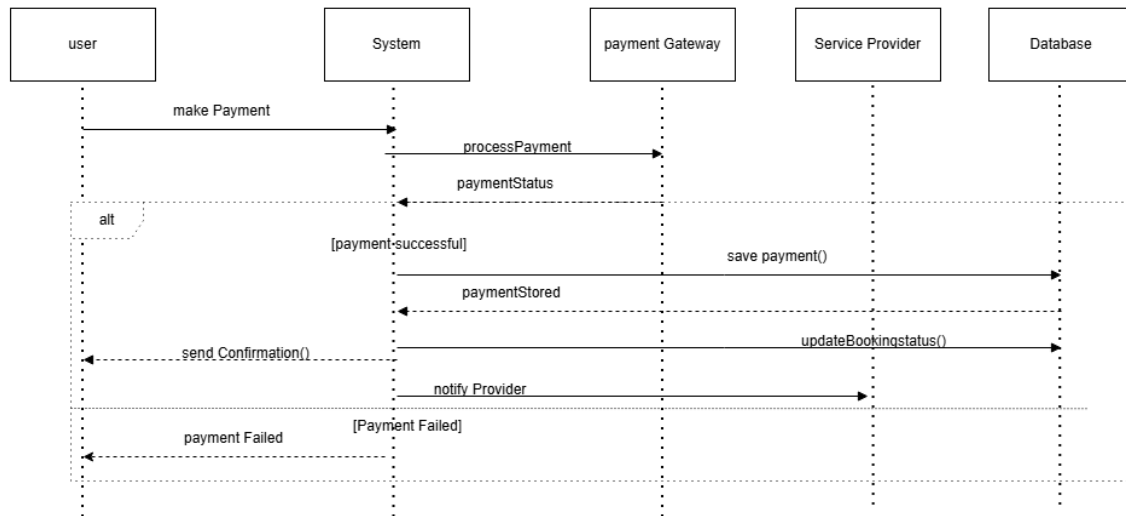


Figure 11 Payment Processing Sequence Diagram

5. Messaging Sequence Diagram:

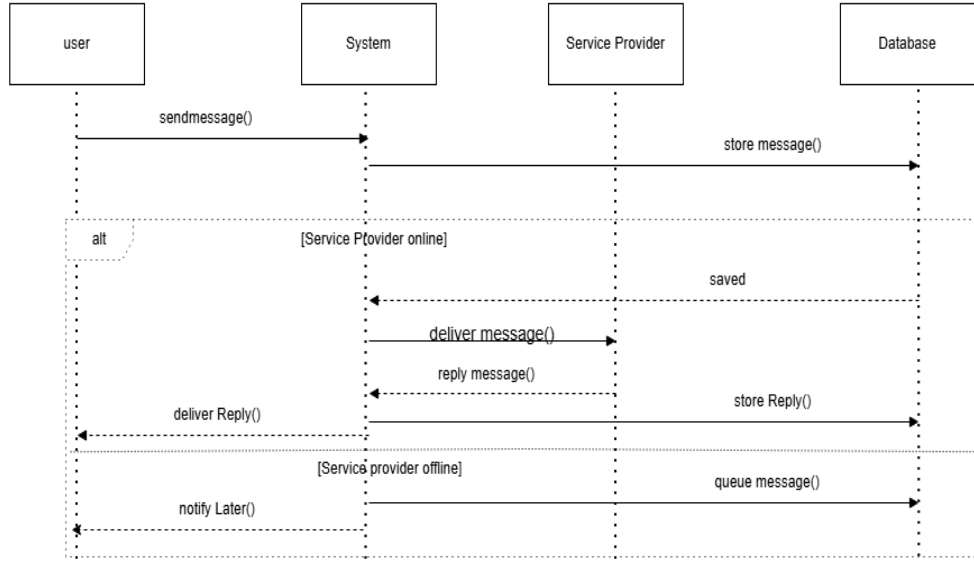


Figure 12 Messaging Sequence Diagram

6. Service Completion & History Sequence Diagram:

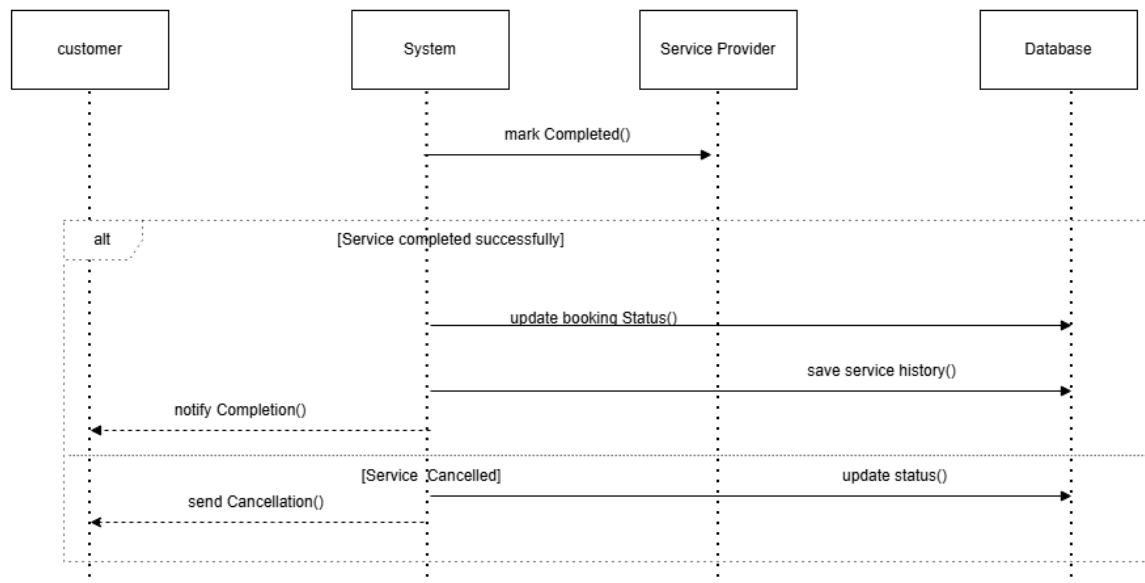


Figure 13 Service Completion & History Sequence Diagram

4.4.3 State Transition Diagram:

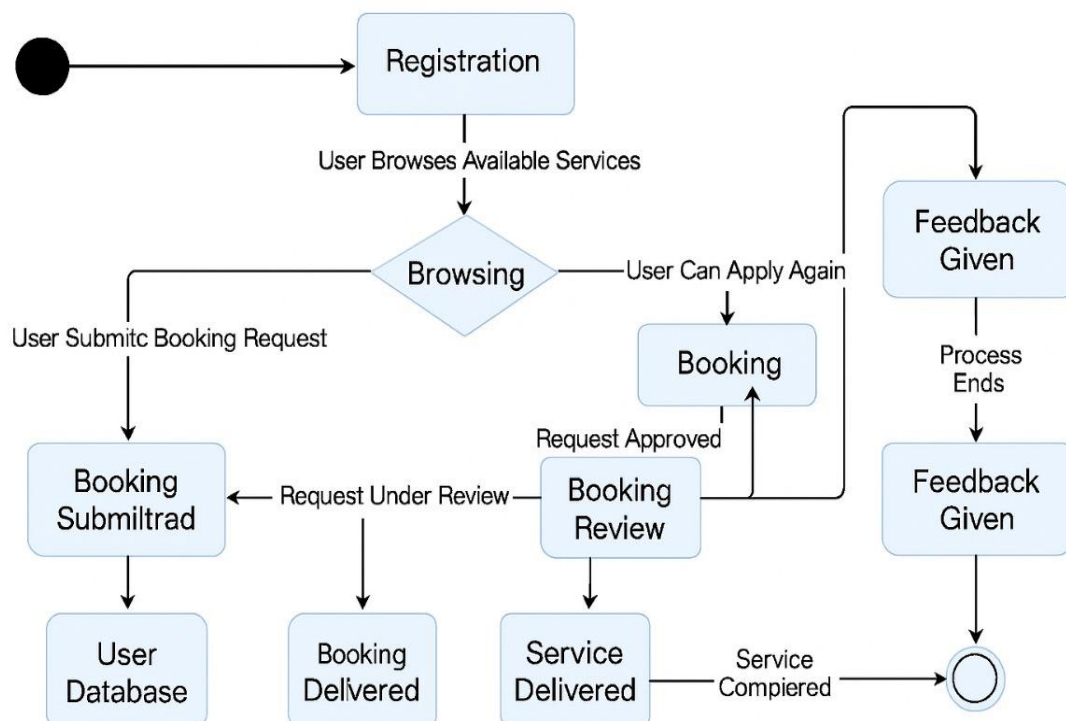


Figure 14 State Transition Diagram

4.5 Architectural Design:

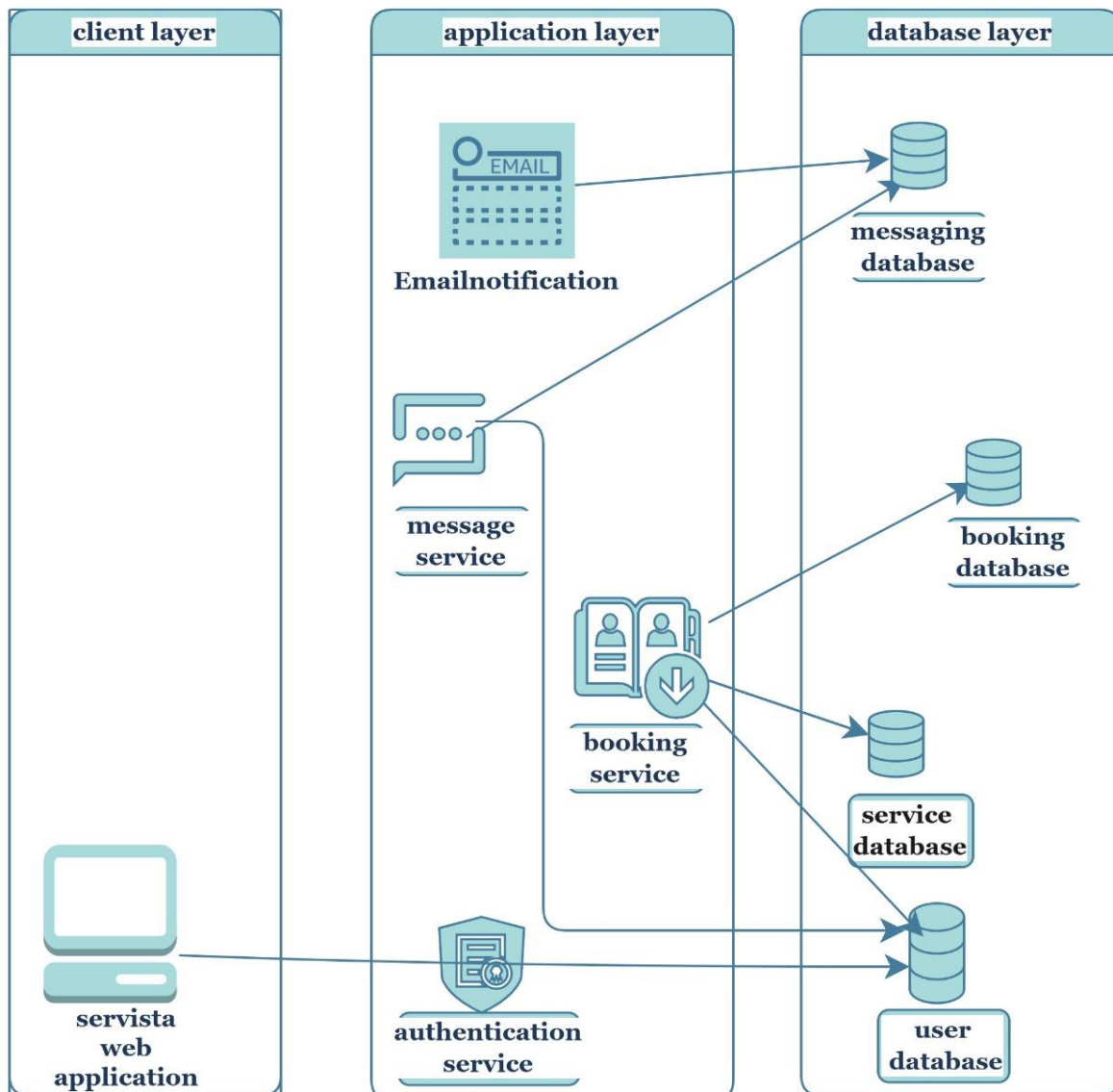


Figure 15 Architectural Design

Our system follows the MVC (Model-View-Controller) architecture using the MERN stack. It is a web based client-server model where the frontend (React) communicates with the backend (Node.js + Express), and the backend interacts with the database (MongoDB).

Main Components:

1. Model:

- Manages data and database operations.
- Includes user data, bookings and service records.

2. View:

- The React-based user interface seen by customer and service provider.
- Displays forms, dashboards and chatbot interface.

3. Controller:

- Handles user requests and connects the view with the model.
- Includes API routes and backend logic written in Node.js.

Component Relationships:

- **Frontend (React)** sends API requests to the backend
- **Backend (Node + Express)** processes requests and accesses MongoDB
- **Database (MongoDB)** stores all data like users, booking and chat history
- **Chatbot logic** is handled in the backend with a rule based system.

UML Component Diagram:

Displays how the frontend, backend, and database interact. Each major function (login, booking, chatbot) is shown as a component connected to others.

4.6 Data Design

The system uses MongoDB as the database to store and manage all data. MongoDB is a NoSQL database, which stores data in the form of collections and documents (JSON format).

Data is stored in flexible, JSON like documents within collections, allowing easy integration within collections, allowing easy integration with the Node.js/Express backend and React Frontend. Each major part of the system has its own collection to keep data organized and easy to access.

Main Collections (Entities):

1.Users Collection:

Stores user details (name, email, phone number, address)

- Stores login credentials (hashed password, role: customer/service provider /admin)
- Support authentication and profile management

2.Services Collection:

- Stores services information (title, description, category, base price, estimated duration)
- Includes service provider references and availability flags
- Support search, filtering, and categorization

3.Bookings Collection:

Stores booking details (customer id, service id, provider id, booking date/time)

Includes status (pending, confirmed, in progress, completed, cancelled)

Stores special requests, total price, and payment status

4.Providers Collection:

Store service provider profiles (business name, contact info, service area)

Includes working hours, ratings, verification status

Links to the services they offer

4.Reviews and ratings Collection:

Stores feedback from customers after service completion

Includes rating (1-5), review text, and timestamps

Linked to booking ID, customer ID, and provide ID

6.Payments Collection:

- Stores transaction details (amount, method, status, transaction ID)
- Links to a specific booking
- Supports payment history and invoice generation

How data is used:

- When a user signs up or logs in, credentials are verified against the user's collection, and JWT token is generated for authenticated sessions.
- When a customer searches for services, data is fetched from the services collection, filtered by category, price, or provider.
- On booking a service, data is added to the bookings collection, and the service status is updated to “pending or confirmed”.
- After service completion, the provider marks the booking as completed, which triggers
 - A payment record is created in the payments collection.

- The customer can add a review, which is stored in the reviews collection.
- Providers can view their assigned bookings, update availability, and manage services through the services and bookings collections.
- Admins can monitor all users, bookings, and payments, and resolve disputes using data from all collections.

4.6.1. Data Dictionary

Purpose:

- To provide a clear and well organized list of all system objects, their attributes, and key data elements.
- Each item includes its data type and a short description of its purpose in the system.

Format:

- **Object Name:** Description of the object's role.
- **Attributes:** All attributes with data types and short descriptions.
- **Methods:** All related methods with parameters and their purpose.

Object: User

Description:

Handles authentication and profile management for all system users (customers, service providers, admins).

Attributes:

- **User id (string):** unique id for the user.
- **Name (string):** Full name of the user.
- **Email(string):** Email address (used for login).
- **Password (string):** hashed password.
- **Phone (string):** Contact number.
- **Address(string):** Physical address.
- **Role(string):** User type (customer, provider, admin).
- **Create At (Date):** Account creation timestamp.

Methods:

- **Register** (name, email, password, role) →Creates a new user account.
- **login** (email, password) →Authenticates user and returns a JWT token.
- **get Profile** (user ID) →Retrieves user details.

Object: Service

Description:

Stores information about services offered by providers.

Attributes:

- **Service id (string):** Unique ID for the service.
- **Title (string):** Name of the service description.
- **Description(string):** Detailed service description.
- **Category (string):** Service category (e.g., cleaning, plumbing, IT).
- **Base Price (Number):** Starting price for the service.
- **Estimate Duration(string):** Expected time to complete (e.g., 2 hours).
- **Provider ID (string):** ID of the service provider offering this service.
- **Is Available (Boolean):** Whether the service is currently bookable.

Methods:

- **add service** (provider ID, title, description, price) → Adds a new service.
- **get service By Category**(category)→returns all services in a category.
- **update Service** (service Id, updates) →modifies service details.

Object: Booking

Description:

Handles booking details between a customer and a service provider.

Attributes:

- **Booking ID (string):** Unique ID for the booking.
- **Customer ID (string):** ID of the customer.
- **Provider ID (string):** ID of the service provider.
- **Service ID (string):** ID of the booked service.
- **Booking Date (Date):** Date when the booking was made.
- **schedule Date (Date):** Date of service delivery.
- **schedule Time (String):** Time of service delivery.
- **status(string):** status (pending, confirmed, in-progress, completed, cancelled).
- **total Price (Number):** Final price after any adjustments.
- **Special Requests (String):** Any customer notes.
- **Methods:**
- **Create Booking (customer ID, service ID, schedule Date, time)** → Creates a new booking.
- **cancel Booking** (booking ID) →Updates status to cancel.
- **Get Bookings by Customer** (customer ID) →Returns all bookings for a customer.
- **Update Booking Status** (booking ID, status) →Changes booking status.

Object: Review

- **Description:**
- Stores customer feedback and ratings for completed services.

- **Attributes:**
- **Review ID (string):** Unique ID for the review.
- **Booking ID (string):** ID of the completed booking.
- **Customer ID (string):** ID of the customer writing the review.
- **Provider ID (string):** ID of the service provider being reviewed.
- **rating(number):** Date when the booking was made.
- **comment(string):** Written feedback.
- **Create At (Date):** Timestamp of review submission.
- **Methods:**
- **add Review (booking ID, rating, comment)** → Creates a new review.
- **Get provide Rating (provider ID)** → Calculates average rating for a provider.
- **Get Review by Provider (provider ID)** → Returns all reviews for a provider.

Object: Payment

- **Description:**
- Handles transaction details for bookings.
- **Attributes:**
- **Payment ID (string):** Unique ID for the payment.
- **Booking ID (string):** ID of the associated booking.
- **amount(number):** payment amount.
- **method(string):** payment method (credit card, cash, wallet).
- **Status(string):** payment status (pending, completed, failed, refunded).
- **Transaction ID (string):** gateway transaction reference.
- **Payment Date (Date):** Timestamp of payment completion.
- **Methods:**
- **process payment (booking ID, amount, method)** → initiates payment.
- **Get payment Status (payment ID)** → Returns current payment status.
- **refund payment (payment ID)** → Processes a refund for cancelled bookings.

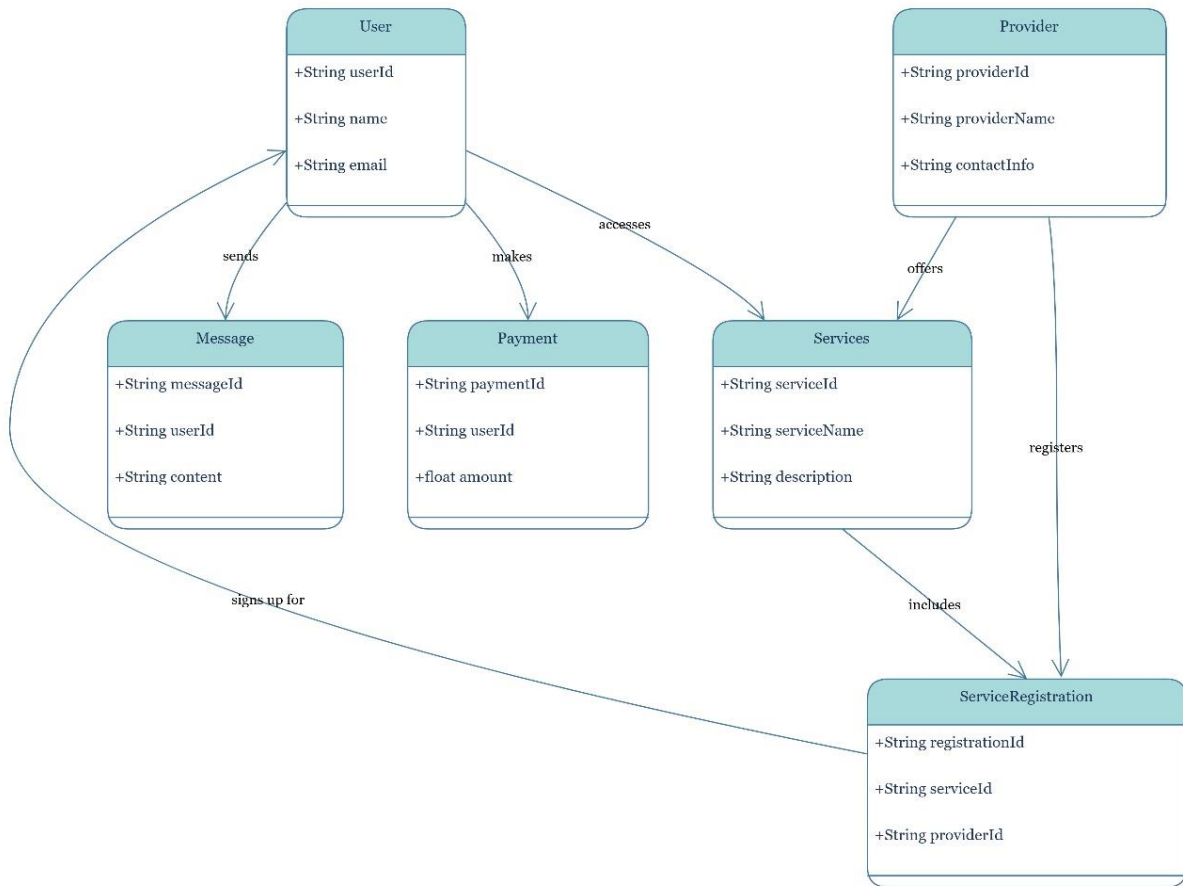


Figure 16 Alphabetical List of System Entities

4.6 User Interface Design:

The Servista web application is designed to be easy to use, intuitive and accessible to users who seek many different services at the same time, communicate with professionals and book them for services. The interface guarantees a perfect user experience with clear navigation, feedback messages, and interactive elements.

General Description of the User Interface:

The system is divided into three main user roles:

- 1) **Administrator:** Supervises users, services and system configuration.
- 2) **Customers:** User seeking the services that they need.
- 3) **Professionals:** Provide the user requires services.

4.6.1 Functionality and user interaction:

1.Home page (destination page)

- Outstanding samples, upcoming discount events.
- Provide login/registration buttons for new users and return them.
- Search bar for required services.

2.User authentication (login/registration)

- Users can register by email/password or social networks.
- Comments: Successful messages or errors in sending the form.

3.Search and filtering Services

- When clicking on any service, it opens details about it.

4.Service booking page

- Show information about service i.e. discounts on the service, description.
- The “Request for Services” button for interested users.
- Comment: Confirmation message.

5.Messaging system

- Customers can communicate with Professional according to services directly.
- Input tray function for chat in real time.
- Comments: Message sent confirmation.

6.Pasure booking application process

- Users can see their booking requests sent.
- Review approved or reject applications.
- Comments: Users receive notifications about the status of the application.

7.Payment and review

- Check the online transaction that you have made.
- rate and review the professionals.

8.User dashboard

- View past services and booking and scheduled services.
- View update and notifications of services.

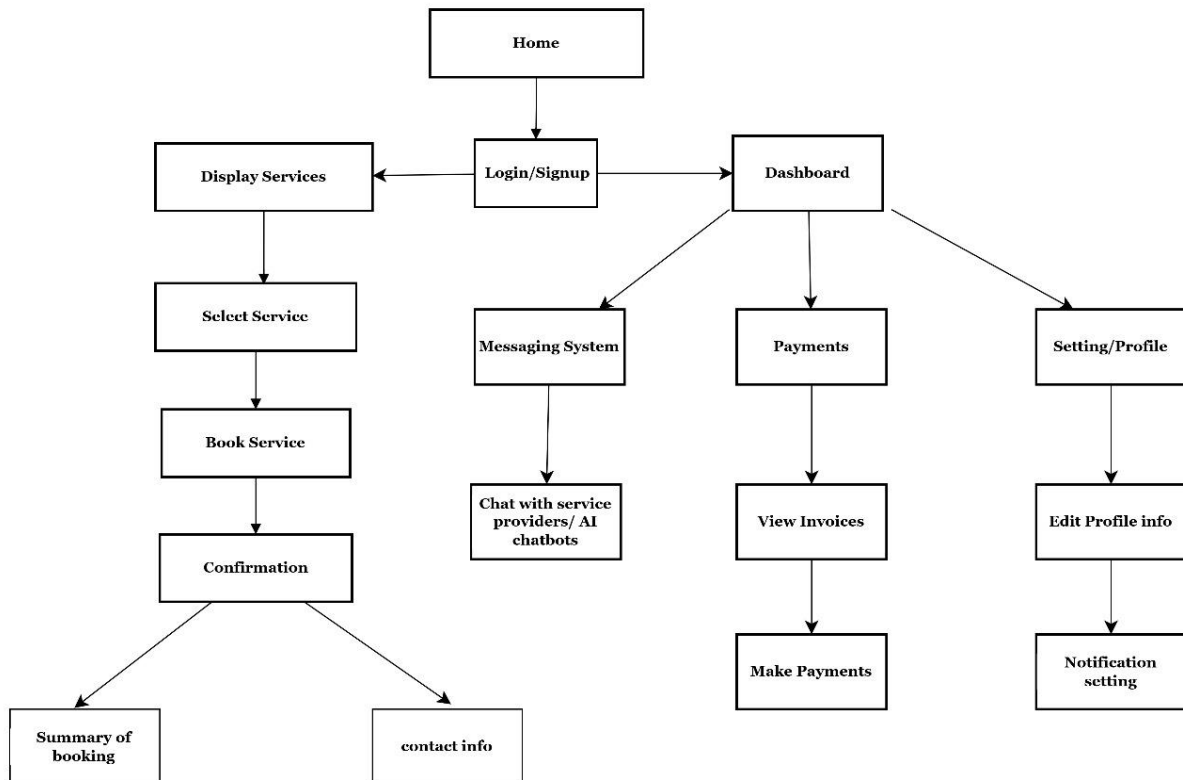
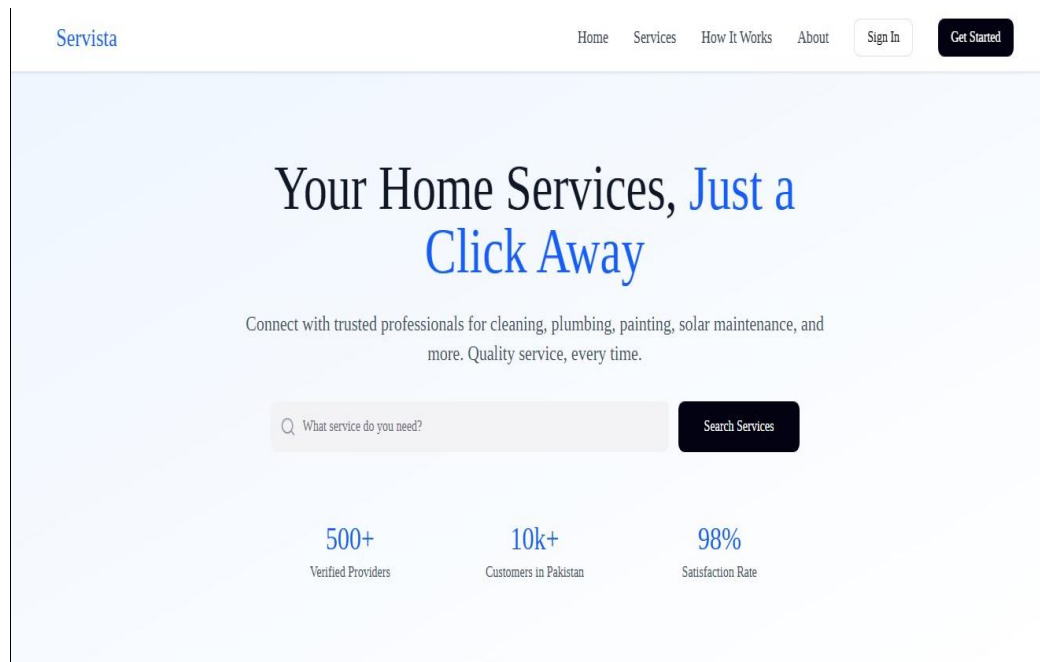


Figure 17 Functionality and user interaction

1.Home Page:



2.Login page:

Servista
Welcome back
Sign in to your service provider account.

Email address

Password

☐ Remember me [Forgot password?](#)

Sign in

Or continue with

Don't have an account? [Sign up for free](#)

Connect with the Best Service Providers

Servista helps you manage your services, connect with clients, and grow your business with powerful tools designed for professionals.

- ✓ Manage bookings and appointments seamlessly
- ✓ Track payments and invoices in one place
- ✓ Build your reputation with verified reviews

10K+ Providers **50K+** Bookings **4.9** Rating

3. Sign Up

Servista
Create your account
Join thousands of service providers

Full Name

Email address

Business Name

Password

Confirm Password

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Create Account

Or sign up with

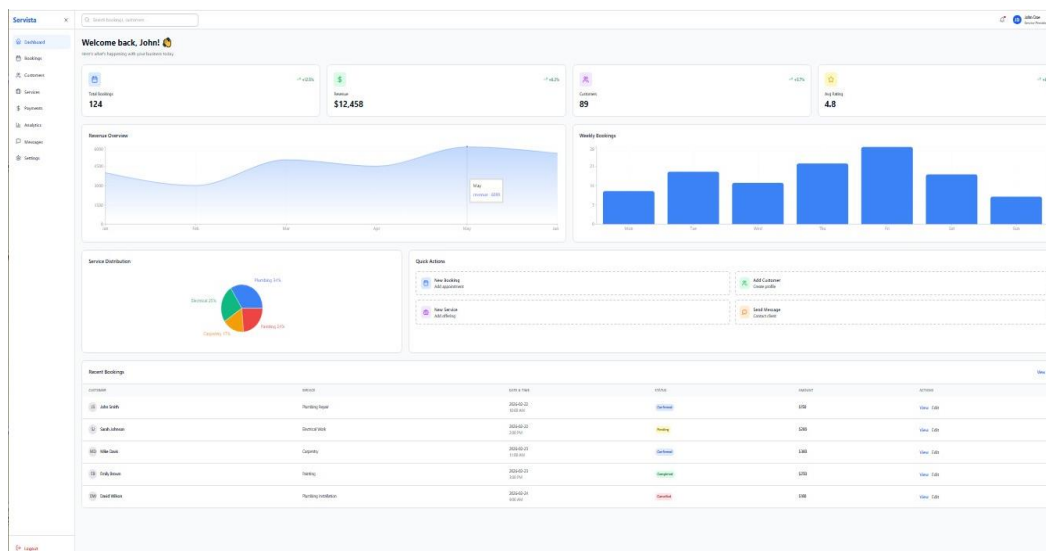
Connect with the Best Service Providers

Servista helps you manage your services, connect with clients, and grow your business with powerful tools designed for professionals.

- ✓ Manage bookings and appointments seamlessly
- ✓ Track payments and invoices in one place
- ✓ Build your reputation with verified reviews

10K+ Providers **50K+** Bookings **4.9** Rating

4. Dashboard:



5.Your NEEDS our SERVICES:

Servista

[Home](#)
[Services](#)
[How It Works](#)
[About](#)
[Sign In](#)
[Get Started](#)

Our Services

From routine maintenance to specialized services, we've got you covered

Cleaning Services
Professional home and office cleaning with eco-friendly products

Plumbing
Expert plumbing repairs, installations, and maintenance

Painting
Interior and exterior painting services by skilled painters

Solar Maintenance
Solar panel cleaning, inspection, and maintenance services

Home Maintenance
General repairs and maintenance for your home

Electrical Services
Licensed electricians for all your electrical needs

Carpentry
Custom woodwork and carpentry solutions

HVAC Services
Heating, ventilation, and air conditioning experts

6.Top rated professionals:

Servista

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Featured Service Providers

Top-rated professionals ready to help you

Premium Cleaning Co.
[Cleaning Services](#)
★ 4.9 (234 reviews)
📍 Karachi, Sindh
🕒 Within 2 hours
From Rs. 1,500/hr [Book Now](#)

Expert Plumbers Inc.
[Plumbing](#)
★ 4.8 (189 reviews)
📍 Lahore, Punjab
🕒 Within 1 hour
From Rs. 2,000/hr [Book Now](#)

Color Masters
[Painting](#)
★ 5 (145 reviews)
📍 Islamabad
🕒 Within 3 hours
From Rs. 1,200/hr [Book Now](#)

Solar Solutions
[Solar Maintenance](#)
★ 4.7 (98 reviews)
📍 Rawalpindi, Punjab
🕒 Same day
From Rs. 3,500/service [Book Now](#)

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Home Fix Experts
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★ 4.8 (167 reviews)
📍 Faisalabad, Punjab
🕒 Within 4 hours
From Rs. 1,800/hr [Book Now](#)

PowerTech Electricians
[Electrical Services](#)
★ 4.9 (203 reviews)
📍 Multan, Punjab
🕒 Emergency 24/7
From Rs. 2,200/hr [Book Now](#)

Sialkot Carpentry Works
[Carpentry](#)
★ 4.8 (156 reviews)
📍 Sialkot, Punjab
🕒 Within 2 hours
From Rs. 1,600/hr [Book Now](#)

7. Your path to Assistance:

[Home](#)
[Services](#)
[How It Works](#)
[About](#)
[Sign In](#)
[Get Started](#)

How It Works

Getting the help you need is simple and straightforward

- Search for Service**
Browse our wide range of home services and find what you need
- Choose Provider**
Compare ratings, reviews, and prices to select the best professional
- Book Appointment**
Schedule a convenient time that works for you
- Get It Done**
Enjoy quality service from verified and trusted professionals

8. Specifications:

[Home](#)
[Services](#)
[How It Works](#)
[About](#)
[Sign In](#)
[Get Started](#)

Why Choose Servista?

Pakistan's most trusted home services platform

- Verified Professionals**
All service providers are thoroughly vetted and background-checked
- 24/7 Support**
Round-the-clock customer support for your convenience
- Quality Guarantee**
100% satisfaction guaranteed or your money back
- Trusted by Thousands**
Join 10,000+ satisfied customers across Pakistan

9. Reviews and Feedback:

Servista

HomeServicesHow It WorksAboutSign InGet Started

What Our Customers Say

Real reviews from satisfied customers across Pakistan

★★★★★

"Excellent service! The cleaning team was professional and thorough. Highly recommend Servista for anyone in Karachi looking for reliable home services."

Ahmed Khan
Karachi
Cleaning Services

★★★★★

"Found the perfect plumber through Servista. Quick response, fair pricing, and quality work. This platform is a game-changer for homeowners!"

Fatima Malik
Lahore
Plumbing

★★★★★

"Great experience with the painting service. The painters were skilled and completed the job on time. Very satisfied with the results."

Hassan Ali
Islamabad
Painting

★★★★★

"Solar panel maintenance was done professionally. The technician was knowledgeable and explained everything clearly. Five stars!"

Ayesha Siddiqui
Rawalpindi
Solar Maintenance

★★★★★

"Needed urgent electrical repairs and Servista connected me with an electrician within an hour. Excellent service and very reliable."

Usman Raza
Faisalabad
Electrical Services

★★★★★

"Very happy with the home maintenance service. Professional team and reasonable prices. Will definitely use Servista again!"

Zainab Ahmed
Multan
Home Maintenance

10. Locations:

Servista

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Serving Cities Across Pakistan

Quality home services available in major cities nationwide

Karachi
Sindh

Lahore
Punjab

Islamabad
Federal Capital

Rawalpindi
Punjab

Faisalabad
Punjab

Multan
Punjab

Peshawar
Khyber Pakhtunkhwa

Quetta
Balochistan

Sialkot
Punjab

Gujranwala
Punjab

Hyderabad
Sindh

Bahawalpur
Punjab

Can't find your city? We're expanding! Contact us to request service in your area.

11. footer:

Servista

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Can't find your city? We're expanding! Contact us to request service in your area.

Servista

Your trusted platform for all home services. Quality, reliability, and convenience in one place.

Services

Cleaning
Plumbing
Painting
Solar Maintenance
Home Maintenance

Company

About Us
Careers
Become a Provider
Contact
Blog

Support

Help Center
Safety
Terms of Service
Privacy Policy

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[f](#)[t](#)[i](#)[in](#)

4.6.2. Screen Objects and Actions

Use Case 1: book a Service (customer side)

Screen Object	Action Performed
Welcome Message	Greet customer name and show service dashboard overview.
Active Booking Section	Displays upcoming or ongoing service bookings.
Booking Status indicator	Shows number of active, completed, and cancelled bookings.
Recommended Services Section	Shows personalized service recommendations based on past bookings.
Service Card	Shows service title, provider name, price, estimation duration, and rating.
View Details Button	Opens full service details including description, provider info, and availability.
Book Now button	Initiates the booking process for the selected service.
Side Menu-Dashboard	Bring customers back to the main dashboard view.
Side Menu-Find Services	Navigates to service search and booking screen.
Side Menu-My Bookings	Opens customer's booking history and upcoming appointments.
Side Menu-Payments	Shows payment history and pending transactions.
Side Menu-Profile	Opens customer profiles for editing personal details.
Slide Menu-logout	Logs the customer out of the system.

Use case 2: Manage Services (Provider Side)

Screen Object	Action Performed
Welcome Message	Displays personalized greetings with the provider's business name.
Total Bookings Counter	Shows the total number of bookings received (today/week/all time).
Pending Counter	Shows number of pending service requests awaiting confirmation.
Completed Counter	Shows a number of successfully completed services.
Cancelled Counter	Shows the number of cancelled bookings.
Service Queue section	Lists upcoming service requests with customer name, scheduled date/time, and status.
Confirm booking button	Accepts a pending booking request.
Mark as completed button	Changes booking status to completed after service delivery.
Add new service button	Allows provider to add a new service offering (title, price, description)
Manage availability button	Updates working hours and service availability schedule.
Side Menu –reviews	Shows customer reviews and ratings received.
Side menu- earnings	Displays earnings summary and payment history.

Use case 3: Admin dashboard (Admin side)

Screen Object	Action Performed
Welcome Message	Displays admin name and system overview.
Total users counter	Shows total registered customers+ providers.
Total bookings counter	Shows system –wide total bookings.
Total Revenue Counter	Displays overall platform revenue.
Pending approvals counter	Shows number of new provider registers awaiting verification.
User management table	Lists all users with options to view, edit, suspend, or delete.
Verify provider button	Approves a newly registered service provider.
Booking management table	Lists all bookings with filtering status, date, or provider.
Dispute Resolution Section	Shows flagged bookings or customer complaints.
Generate report button	Exports system usage or revenue report (pdf/csv).
Side menu-manage categories	Add/ edits/deletes service categories.
Side menu-System Settings	Configures platform settings (commission rates, etc.)

Table 32 Screen Objects and Actions

4.7 Design Decisions:

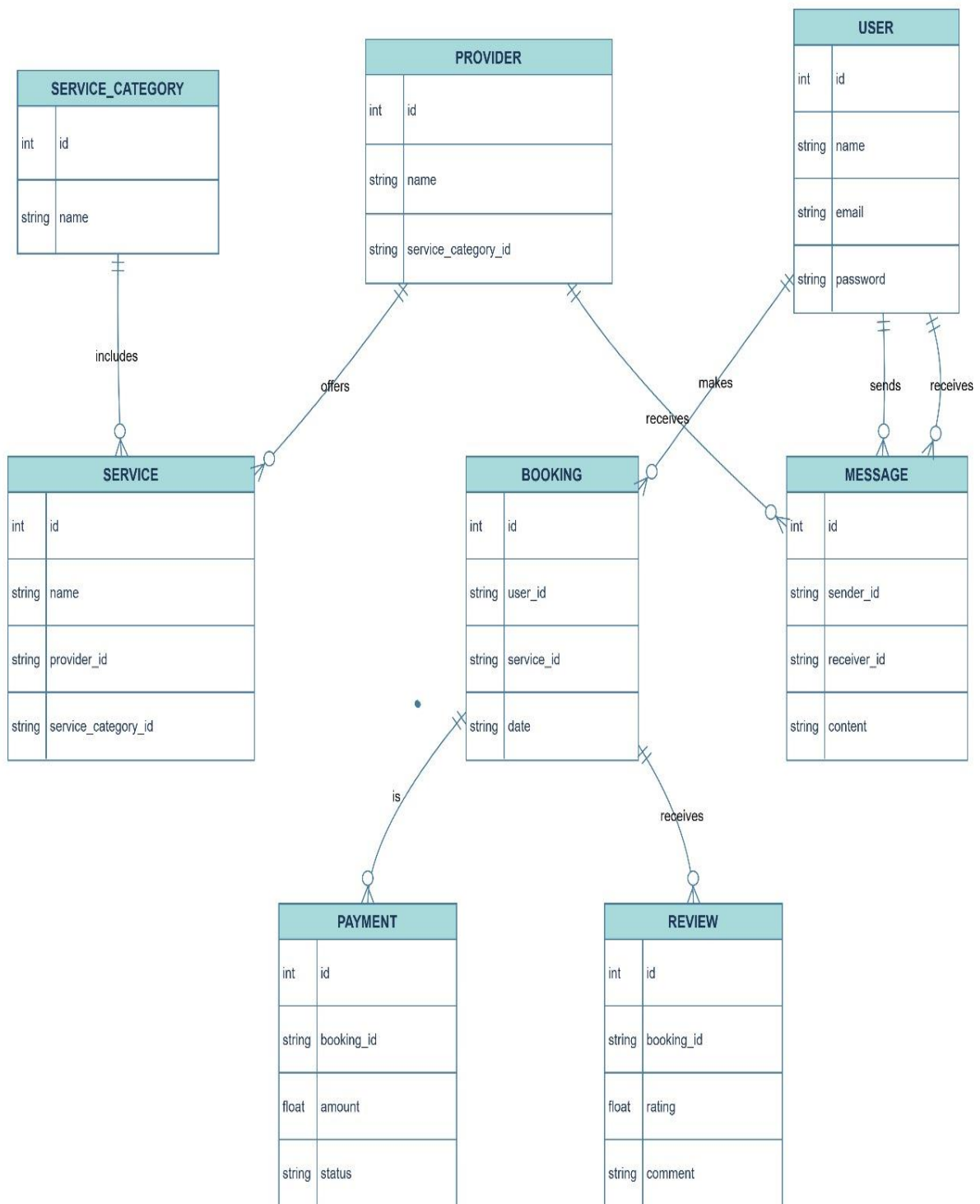


Figure 18 Design Decisions

Design Deisions :

While designing our system, we made the following key decisions to ensure simplicity, maintainability, and performance:

1. Object-Oriented Design Approach

We used the object-oriented (OO) design pattern because it helps in organizing code clearly using objects like User, Service, booking, payment, and review. This makes the system easier to manage, reuse, and extend in the future.

Key benefits for Servista:

- **Reusability:** Core objects (e.g., User, Booking) can be reused across customer, provider, and admin modules.
- **Maintainability:** Changes to one object (e.g., adding a discount field to booking) don't break unrelated parts.
- **Scalability:** New service types or user roles can be added by extending existing objects.

2. MVC Architecture (MERN Stack)

We followed the MVC (Model –view-controller)

Pattern using MongoDB, Express.js, React.js, and Node.js.

- **Model (MongoDB + Mongoose):** Defines schemas for users, Services, Bookings, payments, and reviews. Handles all database interactions and business logic validations.
- **View (React.js):** Provides a dynamic, single page interface for customers, providers, and admins. Uses reusable components (Service card, Booking List, Dashboard Stats) and state management (Context API /Redux).
- **Controller (Express.js +Node.js):** Contains API route handlers that process requests, interact with models, and send responses. Keep business logic separate from UI and database.

Why MVC for Servista:

- **Separation of concerns:** UI changes don't affect database logic, and vice versa
- **Faster development:** Frontend and backend can be built in parallel.
- **RESTful API ready:** Controllers expose clean endpoints for React to consume via Axios/Fetch.

3. Database Choice: MongoDB over SQL

We choose MongoDB (No SQL) instead of a relation database (SQL) for the following reasons:

Factor	MongoDB (Chosen)
Schema flexibility	Dynamic schemas –easy to add fields like special Requests or discount Code anytime
Data structure	Stores nested data naturally (e.g., booking contain service + customer details in one document)
Scalability	Horizontal scaling via sharding – ideal for growing user base
Development speed	JSON-like documents match JavaScript/Node.js ecosystem

Table 33 Database Choice: MongoDB over SQL

For Servista specifically: service listings, bookings, and reviews have varying attributes (e.g., plumbing vs. cleaning services have different fields). MongoDB handles this without complex table redesigns.

4. JWT-Based Authentication

We implemented JSON Web Token (JWT) authentication for secure, stateless user sessions.

- **How it works:** After logging in, the server issues a signed JWT. The react frontend stores it (HTTP Only cookie or local Storage) and sends it with every authenticated request.
- **Why JWT:** No server-side session storage needed, works seamlessly with scalable Node.js Backends, and supports role based access (customer, provider, admin).

5. Role based Access Control (RBAC)

We designed three distinct user roles with specific permissions:

Role	Permissions
Customer	Browse services, book appointments, view booking history, add reviews, make payments
Provider	Manage own services, confirm /cancel bookings, update availability, view earnings and reviews
Admin	Approve providers, manage all users and bookings, resolve disputes, generate platform reports

Table 34 Role based Access Control (RBAC)

Why RBAC: Prevents unauthorized access (e.g., a customer cannot modify provider earnings).

Each role sees only relevant UI components in react, and backend middleware verifies permissions on every API call.

6. Real-Time updates (optional Enhancement)

For future scalability, we designed the system to support WebSocket (Socket.io) integration, enabling:

- Live booking status updates for providers
- Real time notifications for customers when a booking is confirmed
- Queue management for on demand services

4.8 Summary

This chapter explained how the servista system is designed to solve the main problems faced by customers and service providers in the service booking process. It covered the structure of the system, including how different modules (customer, provider, admin) work together and how the data is managed.

The diagrams helped us understand the flow of actions and relationships between components, such as:

- How a customer searches, books and pays for a service
- How a provider manages bookings and earnings
- How an admin oversees the entire platform

We also discussed the assumptions, risks, and limitations we considered while designing the system. Important design choices like using the object –oriented approach and the MERN stack (MongoDB, Express.js, React.js, Node.js) were also explained.

Overall, this chapter played an important role in shaping the project according to the main goals we set at the beginning: providing a simple, scalable and user friendly service booking platform.

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5. **Lee, C., Kim, S. & Adams, T. (2022):** *Ethical Service Delivery in the Digital Age*. Discusses fairness, inclusivity, and responsible platform design.
6. **Miller, K., Brown, D. & Rogers, H. (2021):** *AI for Smarter Service Matching*. Explores how AI can help users find the right professional faster.
7. **Smith, R. & Johnson, M. (2020):** *Digital Transformation in Local Services*. Looks at how traditional services are evolving through tech.
8. **Wilson, P. & Green, B. (2021):** *What Makes a Home Service Platform Work*. A comparative study of successful platforms.
9. **Taylor, S. & Khan, M. (2020):** *Cybersecurity for Service Platforms*. Focuses on keeping user data safe and preventing fraud.

2. Trusted Web Sources

These online articles and reports offer practical insights into running a safe, user-friendly service platform:

10. **Wilson, D. & Green, S. (2021):** *How to Prevent Scams in Online Service Booking*. Published on DigitalSafetyHub.org
11. **Ahmed, R. & Iqbal, T. (2020):** *Post-Service Support: Why It Matters*. From HomeCareInsights.com
12. **Khan, L. & Hussain, M. (2022):** *Making Home Services Accessible for All*. A guide to inclusive design on ServiceDesignToday.com

3. Conference Papers & Presentations

These papers share expert insights from industry events and research forums:

13. **Anderson, J. (2020):** *Challenges in Building All-in-One Service Platforms*. Presented at the Smart Cities Conference.
14. **Brown, K. & Taylor, M. (2019):** *User Experience in Home Maintenance Apps*. Shared at the UX for Services Summit.
15. **Carter, L. & Hayes, R. (2018):** *Ethical Design for Service Platforms*. From the Digital Inclusion Forum.
16. **Garcia, P., et al. (2021):** *AI in Home Services: Smarter Scheduling and Matching*. Presented at the Tech for Homes Symposium.
17. **Lee, T., et al. (2022):** *Communication Tools That Build Trust in Service Apps*. From the Human-Tech Conference.
18. **Miller, R., et al. (2021):** *Digital Transformation of Local Services*. Shared at the Urban Innovation Expo.
19. **Nawaz, H. & Fatima, S. (2023):** *Scaling Service Platforms in Emerging Markets*. Presented at the Pakistan Tech Futures Forum.

4. Other Sources:

- **ChatGPT** (for explanation & diagram ideas)
- **W3School** (quick concepts),
- **GeekforGeeks** (logic & examples)
- **TutorialsPoint** (theory and examples)
- **Library FYP samples** (past students' reports)

5. Online Platforms & Service Models

- UrbanClap Technologies India Pvt. Ltd., "Urban Company Book Home Services," Urban Company, [online]. Available: <https://www.urbancompany.com/> [Accessed: 20-Oct-2025].
- TaskRabbit, Inc., "Task Rabbit Handyman, Cleaning & More," TaskRabbit, [online]. Available: <https://www.taskrabbit.com/> [Accessed: 08-Dec-2025].
- Mahir Company, "Mahir Home Services in Pakistan," Mahir, [online]. Available: <https://mahircompany.com/> [Accessed: 24-Nov-2025].
- Mr. Mahir, "Mr. Mahir Book Trusted Professionals," Mr. Mahir, [Online]. Available: <https://mrmahir.com/> [Accessed: 05-Dec-2025].
- Helpp, "Helpp On-Demand Home Services," Help, [online]. Available: <https://helpp.pk/> [Accessed: 08-Jan-2026].